A Dynamic, Decomposable Operational Cyber Risk Posture Index

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Abstract

Cybersecurity risk is inherently dynamic, multi-dimensional, and shaped by the continuous interaction between technology, adversary capability, organizational behavior, and dependency structure. While architectures, tooling, and governance frameworks evolve rapidly, the executive questions remain stable: *What is our current risk posture? What is driving it? How exposed are we, how severe would impact be, and how quickly can we detect, respond, and recover?*

This paper introduces the Cyber Risk Posture Index (CRPI), a telemetry-compatible, framework-agnostic, decomposable operational posture index for representing organizational cyber risk as a function of time. The governing model is mathematically specified rather than based on discretionary top-level scoring: threat exposure is combined with resilience-adjusted consequence, operational execution is modulated by observability, and past unresolved exposure is represented by a bounded stock-flow state. In this representation, current residual exposure contributes to a persistent score over time, effective mitigation reduces that score, and the state remains between zero and one. It is a modeled measure of unresolved exposure, not a directly observed vulnerability count. Regularized geometric aggregation prevents a single zero-valued sub-metric from unintentionally collapsing an otherwise non-zero construct. The model produces a bounded, scope-aware operational risk metric together with traceable diagnostics for hazard pressure, mitigation effectiveness, instantaneous residual posture, and persistent unresolved exposure. Subjectivity is not claimed to be eliminated; it is constrained to explicit, versioned composition and calibration parameters. The paper further specifies measurement interfaces, evidence-coverage gates, uncertainty propagation, calibration governance, and implementation-oriented sub-metrics as a reproducible foundation for subsequent empirical validation.

1 Introduction

At the executive and board level, cybersecurity is ultimately a risk management problem. Senior leaders repeatedly ask a small number of fundamental questions: *How risky are we today? What is our current exposure? What is the likelihood of meaningful impact? If something happens, how effectively can we detect it, contain it, and recover?* Despite substantial investment in security programs and tooling, these questions are rarely answered with quantitative clarity that is both trustworthy and traceable.

Most existing approaches to cyber risk assessment rely on static, point-in-time evaluations, qualitative scoring, control-centric dashboards, or framework-specific maturity models. While such approaches are valuable for compliance, assurance, and control management, they struggle to represent risk as a continuously evolving property of a living socio-technical system. As new technologies emerge and adversaries adapt, models that encode technology-specific assumptions can lose explanatory power or become difficult to compare across time, business units, and organizations [1, 2, 6, 7, 9, 10, 11, 12].

This work proposes a different approach: a small set of stable, mathematically specified equations operating over cyber-risk constructs whose semantics remain useful across different technologies and governance frameworks. The Cyber Risk Posture Index (CRPI) is not intended to estimate breach probability or monetary loss. It is an operational posture index designed to convert heterogeneous evidence into a bounded, decomposable, time-aware representation of current and persistent residual exposure. Mathematical grounding reduces reliance on discretionary top-level weights, while explicit calibration governance makes the remaining judgment visible, local, versioned, and auditable.

The paper makes five specific contributions:

1. it defines a technology-agnostic set of Tier 2 cyber-risk constructs separating observability, consequence, threat exposure, operational execution, and resilience;
2. it specifies a zero-safe regularized aggregation operator that preserves geometric sensitivity to weak components without allowing one zero-valued input to collapse an otherwise non-zero construct;
3. it introduces explicit equations for resilience-adjusted consequence, observability-modulated mitigation, instantaneous residual posture, and bounded stock-flow persistence rather than relying on discretionary top-level weights;
4. it provides measurement, coverage, uncertainty, and governance interfaces for mapping heterogeneous telemetry into stable normalized constructs; and
5. it demonstrates how AI-era changes in public threat signals can be represented without treating temporal correlation as proof that AI produced the observed change.

Organization of the research

The paper first establishes the operational concepts and five measurable dimensions of posture, then explains how their evidence is qualified and how changing conditions can carry effects across reporting periods. With that vocabulary established, it introduces the governing equations, demonstrates their calculation with synthetic examples, and separates mathematical verification from prospective empirical validation. Numbered equations are the formal calculation definitions, and references such as Equation 24 refer to those displayed expressions. Figures and tables illustrate the declared profile rather than validated security thresholds.

The paper is self-contained as a mathematical and methodological specification. Executable reference calculations and non-sensitive fixtures are available at `https://github.com/jmsbooth/cyber-risk-posture-model`, with detailed artifact provenance described in the reproducibility section. Access to the repository is not required to interpret the model, and the repository does not contain an organizational validation dataset.

2 Design Philosophy and Invariance Goals

The CRPI is designed to preserve a stable governing structure while the evidence used to instantiate that structure evolves. Its invariance claim applies to construct semantics and equation form, not to universal calibration coefficients.

- **Technology-Agnostic Governing Structure**: The equations do not depend on a particular product, platform, control catalog, or architecture. New technologies change evidence and calibration, not the role of the governing constructs.

- **Framework Agnosticism**: Inputs may be derived from NIST, ISO, CIS, sector-specific standards, or internal control models without altering the governing equations.
- **Temporal Persistence**: Risk posture is time-dependent. The CRPI models both instantaneous residual exposure and a bounded stock of unresolved exposure that accumulates and dissipates over time.
- **Scope-Aware Interpretability with Decomposability**: The model produces a bounded operational risk metric for component, system, network, domain, and enterprise analysis while retaining traceability to hazard pressure, mitigation effectiveness, persistent exposure, Tier 2 constructs, and underlying evidence.
- **Separation of Structure and Instantiation**: Metric definitions, telemetry sources, normalization bounds, smoothing parameters, and calibration parameters may evolve under governance while equation form and construct semantics remain stable.
- **Mathematical Grounding and Governed Subjectivity**: The model does not claim to eliminate expert judgment. It constrains judgment to explicit composition and calibration parameters whose meanings, ranges, versions, and evidence bases are auditable.
- **Evidence Qualification**: A posture index must be accompanied by evidence coverage and, where supported by the underlying measurements, propagated uncertainty so that sparse or weak evidence cannot be mistaken for high-certainty assessment.

These principles are intended to preserve longitudinal interpretability without claiming that cyber risk follows immutable physical laws. The objective is a stable mathematical interface whose assumptions and calibration choices remain explicit.

3 What the CRPI Is and Is Not

The CRPI is intended to be a quantitative operational cyber-risk posture model, not a replacement for governance, assurance, scenario analysis, or financial cyber-risk quantification.

The CRPI **is**:

- a bounded, time-aware operational cyber-risk posture index;
- a decomposition of posture into stable underlying constructs and state;
- a scope-aware diagnostic supporting engineering, operations, model governance, and executive views;
- a telemetry-compatible interface for mapping heterogeneous evidence into a common representation; and
- a potential upstream signal for downstream scenario-based or financial risk analysis.

The CRPI **is not**:

- a control framework or compliance regime;
- a maturity model;
- a direct prediction engine for specific attacks;

- a probability of breach;
- a direct financial loss model; or
- a guarantee of organizational security.

This distinction matters because the CRPI is designed to unify and interpret operational evidence, not replace governance, control validation, scenario reasoning, or expert judgment.

4 Conceptual Architecture of CRPI

The Cyber Risk Posture Index (CRPI) distinguishes the conditions that create exposure, the consequences that would follow a compromise, and the capabilities that change how those conditions develop. Its principal contribution is a structured relationship between these distinct questions, not the claim that an observed score is a probability of attack or loss.

The model begins with five Tier 2 constructs. *Threat Exposure (TE)* represents attackability and adversary-relevant conditions. *Consequence Severity (CS)* describes the potential harm to confidentiality, integrity, and availability. *Continuous Monitoring (CM)* describes how reliably the scoped environment is observed. *Security Operations Effectiveness (SOE)* describes the demonstrated ability to detect, respond, contain, remediate, and make changes safely. *Resilience and Risk Assurance (RRAS)* represents recovery, continuity, and assurance over dependencies. A single reporting resilience score can be useful for summaries, but the model also asks which resilience capabilities are relevant to each type of consequence.

Three distinctions govern how these constructs are used. First, observed attack exposure is not itself the same as consequence: an externally reachable system can have low or high mission impact. The model combines them into *hazard pressure*, defined here as a relative score for exposure weighted by consequence after applicable structural resilience. Second, visibility is not the same as action. Monitoring can inform and support operational mitigation, but observing a vulnerability does not fix it. Third, current residual exposure is not a complete account of unresolved history. A newly improved environment may still carry unresolved consequences from an earlier period of weaker controls, incomplete remediation, or elevated attack conditions.

The temporal distinction is represented by two quantities. The *instantaneous residual posture* is the amount of modeled hazard remaining after current mitigation. The *persistent unresolved exposure state* is a bounded stock-like measure that retains part of prior unresolved exposure, increases when residual exposure is supplied, and decreases when mitigation retires that state. The stock is dimensionless and restricted to the interval $[0, 1]$; its accumulation and retirement rates are explicit assumptions to be evaluated against suitable independent evidence. This is a modeling choice, not evidence that organizations exhibit a universal stock-flow law.

The final reported CRPI is the larger of the current residual score and the persistent state. Selecting the larger quantity is an explicit conservative reporting policy: it avoids calling posture low merely because one of the two channels has improved. Both channels and their contributing measurements must accompany the scalar. A change in the scalar can otherwise conceal whether the cause was an exposure surge, a consequence reassessment, improved operational execution, or retained history.

5 Tier 2 Risk Dimensions

The CRPI decomposes cyber risk into five Tier 2 dimensions that correspond to stable, technology-agnostic constructs: Continuous Monitoring (CM), Consequence Severity (CS), Threat Exposure

Table 1: Conceptual roles before the governing equations are introduced

Concept	Operational question
Exposure and consequence	What conditions make adverse action possible, and what harm could follow?
Monitoring and operations	Can the organization reliably identify the condition and act effectively?
Resilience	Which consequences can architecture, recovery, and trusted dependencies actually reduce?
Current residual	What modeled exposure remains under present conditions?
Persistent state	What modeled unresolved exposure is carried across time intervals?
Evidence qualification	What is actually known about the declared scope, and what remains uncertain or undiscovered?

(TE), Security Operations Effectiveness (SOE), and Resilience and Risk Assurance Score (RRAS). Each Tier 2 metric is bounded in $[0, 1]$ and instantiated from measurable sub-metrics derived from telemetry, governance artifacts, engineering workflows, or architectural evidence.

The purpose of the Tier 2 layer is twofold. First, it decomposes cyber risk into constructs that remain semantically meaningful across technology eras, organizational models, and governance frameworks. Second, it provides a stable interface between the top-level CRPI equations and the evolving evidence sources used to operationalize the model in practice.

A central design requirement is that the mathematical role of each Tier 2 metric remain stable even as specific sub-metrics, tooling, or evidence sources change. In other words, the CRPI assumes that organizations will monitor different technologies, process different data, face different threat landscapes, operate with different engineering practices, and depend on different ecosystems, but it maintains that the underlying risk constructs of visibility, consequence, susceptibility, execution capability, and structural survivability remain stable.

Accordingly, the Tier 2 layer is defined by *construct semantics* rather than by tool categories. This distinction is important. A Tier 2 metric should not be read as a bundle of today's technologies, but as an abstract risk function whose current instantiation may be realized using whichever telemetry and governance mechanisms an organization has available. This is what allows the CRPI to remain defensible over time.

Internal Tier 2 weights, where present, are treated as construct-composition parameters rather than discretionary preference levers. They exist to ensure semantic completeness of a construct when multiple sub-metrics are required to operationalize it. These weights are therefore constrained by model governance, versioned, empirically reviewable, and not intended to encode shifting business risk appetite at the level of the top-level CRPI output.

The sections that follow define each Tier 2 metric in terms of its purpose, construct boundaries, mathematical form, representative measurement interfaces, and role within the larger CRPI system.

The measurement tables below make the input contract explicit. A discrete input is a bounded, auditable measure that is normalized before aggregation; it is not required to come from a particular product. Each implementation should record the source, owner, normalization bounds, expected cadence, missing-data policy, and consequence-domain or resilience mapping for every input it uses. The listed inputs are recommended starting points and may be replaced by semantically equivalent measures under the calibration profile.

5.1 Continuous Monitoring (CM)

5.1.1 Purpose and Philosophy

Continuous Monitoring (CM) represents an organization's sustained ability to observe security-relevant system state and to continuously verify that intended controls, instrumentation, and operational policies remain present, active, and current. CM is not defined by any specific tool, but by the enduring properties of visibility coverage, freshness, integrity, and verification across relevant assets and control points.

The foundational role of CM in the CRPI is to model the epistemic quality of the organization's security posture: an organization cannot manage, remediate, or validate risk in near real-time if it cannot reliably observe changes, detect drift, or confirm the operational state of controls. CM therefore measures capability to know rather than capability to respond.

5.1.2 Construct Definition

CM quantifies four stable invariants:

1. **Coverage**: Are all in-scope assets and control points instrumented and reporting?
2. **Freshness**: How recent is the telemetry and validation evidence?
3. **Integrity**: Can the telemetry and control-state evidence be trusted?
4. **Verification**: Does monitoring validate control state rather than merely collect signals?

5.1.3 Metric Form

CM is modeled as

$$CM(t) = \sum_{j=1}^{m} w_j^{CM} \widetilde{CM}_j(t), \qquad \sum_{j=1}^{m} w_j^{CM} = 1, \tag{1}$$

where each $\widetilde{CM}_j(t) \in [0, 1]$. These normalized primitive values represent observation coverage, freshness, integrity, and independently verified control state. The non-negative, sum-to-one weights specify how they are combined into Continuous Monitoring (CM). A high weighted aggregate is not evidence that an omitted source or undiscovered asset was captured. Individual primitive values and source-qualification gates remain necessary.

5.1.4 Measurement Interfaces and Representative Inputs

Representative data sources include authoritative asset inventories, telemetry presence and last-seen indicators, policy and configuration repositories, and audit or review artifacts. The recommended discrete inputs are:

Table 2: Recommended discrete inputs for Continuous Monitoring

Discrete input	Recommended evidence or input	Normalization	Cadence
Telemetry coverage	Fraction of in-scope assets and control points with valid reporting.	Higher is better; asset inventory is the denominator.	Continuous or daily
Telemetry freshness	Age since the last valid observation for each required signal.	Lower age is better; use a bounded decay or threshold map.	Minutes–hours
Telemetry integrity and provenance	Completeness, schema validity, source provenance, and evidence that observations were not silently altered.	Higher is better; document verification tests.	Continuous or daily
Audit-log coverage	Required event classes present, retained, and queryable for in-scope systems.	Higher is better; map required event classes to coverage.	Daily
Control-state drift detection	Agreement between intended control state and observed configuration or behavior.	Higher is better; normalize drift detection coverage and timeliness.	Daily or change-driven
Control update cadence	Timeliness and completeness of required control and policy updates.	Higher is better; use governed age thresholds.	Weekly or change-driven
Procedure freshness	Recency and validation status of operational and response procedures.	Higher is better; use review-age and exercise evidence.	Scheduled or event-driven

5.1.5 Boundary Conditions

CM excludes incident response performance (SOE), exploitability signal and adversary pressure (TE), structural resilience (RRAS), and impact severity (CS).

5.1.6 Interpretation

A high CM score indicates continuous, trustworthy visibility into security-relevant state. A low CM score indicates that posture may be poorly observed or managed in near real-time. Within the governing model, CM modulates operational execution through Equation 24; visibility improves the effectiveness of action but is not itself treated as remediation.

5.2 Consequence Severity (CS)

5.2.1 Purpose and Philosophy

Consequence Severity (CS) quantifies the consequence potential of compromise by measuring the sensitivity, criticality, and mission relevance of information and functions handled by a system, component, or organization. CS is intentionally defined in terms of consequence rather than threat or control mechanism.

5.2.2 Construct Definition

The baseline profile captures consequence through governed domains such as:

1. **Confidentiality and Sensitivity**: intrinsic sensitivity, classification, regulatory exposure, ethical obligation, and derived or inferential sensitivity;
2. **Integrity and Mission Consequence**: harm from unauthorized change, corrupted decisions, unsafe operation, or loss of mission trust; and
3. **Availability and Continuity**: harm resulting from loss or degradation of access to data, systems, or essential services.

Other environments may use a different finite domain set $\mathcal{D}$ when the mapping is documented and governed. The domain structure is retained in the governing hazard calculation because resilience does not mitigate every consequence class equally.

5.2.3 Metric Form

Each consequence domain $D_j(t)$ is composed from normalized sub-metrics assigned only to that domain. The reporting aggregate $CS(t)$ is then calculated using Equation 18. Hazard pressure uses the resilience-adjusted $D_{\text{eff}}(t)$ from Equation 19 rather than applying a single global fragility multiplier to the aggregate CS score.

The reporting Consequence Severity (CS) score combines impact domains without subtracting resilience. In contrast, D_{eff} adjusts each relevant domain only by its applicable resilience projection. An ordinal five-level impact rubric mapped to $0, 0.25, 0.5, 0.75, 1$ assumes equally spaced numeric levels; this is an illustrative normalization to be evaluated against independent consequence anchors.

5.2.4 Measurement Interfaces and Representative Inputs

Representative sources include data inventories and catalogs, classification and labeling artifacts, business impact analyses, lineage graphs, mission dependency models, and access/usage metadata. The recommended discrete inputs are:

Table 3: Recommended discrete inputs for Consequence Severity

Discrete input	Recommended evidence or input	Primary domain	Normalization
Direct data sensitivity	Classification, regulated-data tags, privacy metadata, and ethical sensitivity assigned to information assets.	Confidentiality	Higher sensitivity is higher CS.
Derived and inference sensitivity	Lineage, linkage, re-identification, and inference-amplification assessments.	Confidentiality	Higher inferred harm is higher CS.
Regulatory and ethical consequence	Governed obligations, affected populations, and consequence assessments tied to the scoped entity.	C; I	Higher consequence is higher CS.
Mission and business criticality	Business impact analysis, mission dependency, service criticality, and affected-process inventory.	I; A	Higher criticality is higher CS.
Integrity consequence	Safety, decision, transaction, and mission harm from unauthorized change or corruption.	Integrity	Higher harm is higher CS.
Availability and continuity impact	Service criticality, tolerable downtime, recovery objectives, and essential-function dependencies.	Availability	Higher harm is higher CS.

5.2.5 Boundary Conditions

CS excludes exploitability signal (TE), monitoring quality (CM), response effectiveness (SOE), and resilience capability (RRAS). RRAS affects realized consequence through the explicitly governed R_j and ρ_j interactions in Equation 19.

5.2.6 Interpretation

A high CS score indicates severe aggregate consequence if compromise occurs. A high D_{eff} indicates that substantial consequence remains even after accounting for the forms of harm that structural resilience can demonstrably mitigate.

5.3 Threat Exposure (TE)

5.3.1 Purpose and Philosophy

Threat Exposure (TE) measures an entity's susceptibility to adversarial action given prevailing threat conditions, system characteristics, and exposure properties. TE represents the susceptibility and adversarial-exposure dimension of posture and is intentionally separated from consequence, response capability, and resilience.

5.3.2 Construct Definition

TE captures two interacting invariants:

1. **External Adversary Pressure**: relevance, capability, observed activity, and targeting intensity associated with adversaries applicable to the environment;
2. **Internal Exposure Conditions**: reachable attack surface, exploitability, identity/privilege exposure, and conditions that enable control bypass.

5.3.3 Metric Form

TE uses the zero-safe regularized geometric operator from Equation 16:

$$TE_{\text{inst}}(t) = \mathcal{G}_{\varepsilon_{TE}}\left(\widetilde{X}_1(t), \ldots, \widetilde{X}_q(t); \mathbf{w}^{TE}\right), \tag{2}$$

with temporal smoothing

$$TE(t) = \alpha_{TE} TE_{\text{inst}}(t) + (1 - \alpha_{TE}) TE(t-1). \tag{3}$$

Here TE_{inst} is the current exposure score from normalized, scoped evidence. The smoothing coefficient α_{TE} determines what share of the new assessment is used, with the remainder carried from the prior score. This can suppress noisy fluctuations but also delay reaction to true exposure surges. The sampling cadence, initial value and coefficient must be governed and sensitivity tested. A zero in one valid TE sub-metric therefore creates a multiplicative penalty proportional to its governed construct weight; it does not mathematically assert that all threat exposure is zero.

5.3.4 Measurement Interfaces and Representative Inputs

Representative sources include exposure inventories, vulnerability and exploitability signals, threat-intelligence artifacts, identity and privilege data, and architecture characteristics. The recommended discrete inputs are:

Table 4: Recommended discrete inputs for Threat Exposure

Discrete input	Recommended evidence or input	Normalization	Cadence
External attack surface	Internet exposure, externally reachable services, exposed interfaces, and time exposed.	Higher exposure is higher TE.	Continuous or daily
Internal reachability	Trust relationships, segmentation boundaries, reachable paths, and lateral-movement exposure conditions.	Higher reachability is higher TE.	Daily/change-driven
Exploitability signal	Vulnerability severity, age, exploit evidence, affected exposure, and remediation state. This is an exposure signal, not a breach probability.	Higher signal is higher TE.	Daily/event-driven

Discrete input	Recommended evidence or input	Normalization	Cadence
Identity and privilege exposure	Privileged identities, privilege paths, authentication coverage, and excessive-access conditions.	Higher exposure is higher TE.	Daily
Threat-intelligence relevance	Governed intelligence on applicable adversaries, targeting, observed activity, and environmental relevance.	Higher relevance is higher TE.	Daily/weekly
Control-bypass potential	Exceptions, unsafe paths, compensating-control gaps, and conditions enabling bypass.	Higher potential is higher TE.	Change-driven

5.3.5 AI-Era Threat Acceleration and Model Interpretation

AI does not require a new governing risk dimension. To the extent that AI changes the speed, scale, or cost of reconnaissance, vulnerability discovery, exploit development, social engineering, or defensive evasion, those changes should appear in the same TE evidence channels already represented by the model [25, 26]. The CRPI therefore treats AI as a potential driver of existing exposure variables rather than as an independent top-level risk multiplier.

This paper does *not* infer that observed increases in vulnerability publication, known exploitation, or other cyber activity were produced by AI. Temporal association does not identify the source of an observed change. Public cyber datasets are affected by multiple confounders, including changes in vulnerability reporting, growth in the CVE Numbering Authority (CNA) ecosystem, vendor disclosure practice, CISA collection and catalog policy, security-research activity, and the underlying installed technology base. The purpose of the following illustration is narrower: to show how an externally observed acceleration or change signal would enter TE and propagate through the unchanged CRPI equations.

5.3.6 Public-Data AI-Era Illustration

Figure 1 uses two descriptive public signals. Annual CVE publication volume is normalized as a discovery/reporting-pressure proxy using official CVE Program annual totals [13]. CISA KEV additions are normalized as an observed-exploitation catalog proxy using the CISA KEV `dateAdded` field [15, 28]. Because the KEV catalog launched with substantial historical backfill, only 2023–2025 additions are used in the composite illustration.

Let

$$D_y = \frac{\mathrm{CVE}_y}{\max_y \mathrm{CVE}_y}, \qquad K_y = \frac{\mathrm{KEVAdditions}_y}{\max_y \mathrm{KEVAdditions}_y}. \tag{4}$$

For the three-year composite illustration only, equal weights and $\varepsilon_{TE} = 0.05$ are used:

$$TE_y^{\mathrm{pub}} = \mathcal{G}_{0.05}\left(D_y, K_y; 0.5, 0.5\right), \qquad y \in \{2023, 2024, 2025\}. \tag{5}$$

Holding resilience-adjusted consequence at $D_{\mathrm{eff}} = 0.48$ solely to visualize propagation,

$$H_y^{\mathrm{pub}} = 0.48\, TE_y^{\mathrm{pub}}. \tag{6}$$

The 2022.5 marker in Figure 1 is a contextual marker for the period of widespread generative-AI availability, not a treatment boundary and not evidence that AI produced the observed changes.

Table 5: Public descriptive inputs and derived CRPI illustration

Year	CVEs	D_y	KEV adds	K_y	TE_y^{pub}	H_y^{pub}
2018	16,512	0.342	–	–	–	–
2019	17,308	0.359	–	–	–	–
2020	18,375	0.381	–	–	–	–
2021	20,161	0.418	–	–	–	–
2022	25,059	0.519	–	–	–	–
2023	28,961	0.600	187	0.763	0.677	0.325
2024	40,077	0.831	186	0.759	0.794	0.381
2025	48,244	1.000	245	1.000	1.000	0.480

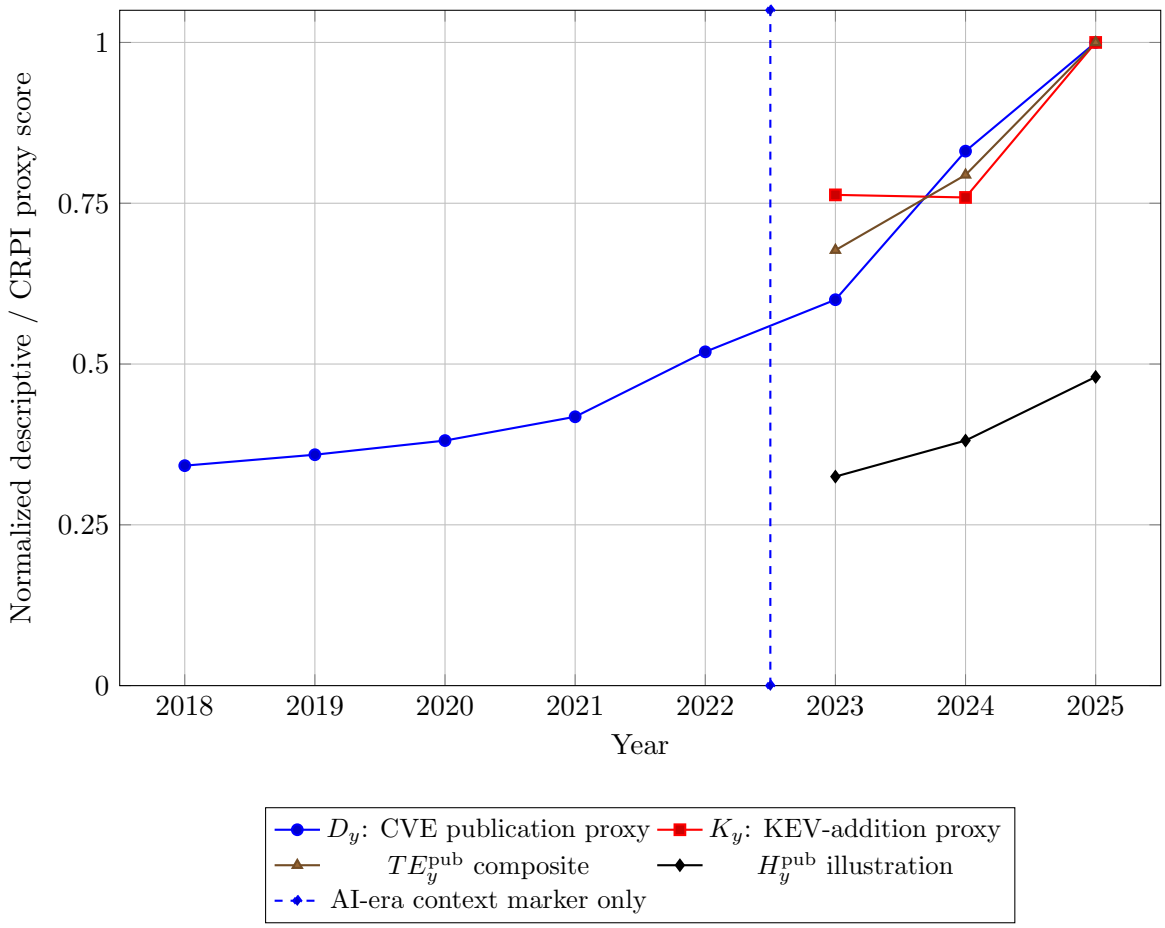


Figure 1: Descriptive public-data illustration of how changing threat signals enter the CRPI calculation path. The CVE series is affected by reporting-system growth and other confounders; KEV is a curated catalog rather than a census of exploitation. The 2022.5 marker is contextual only. The figure does not estimate an AI treatment effect and does not attribute the observed changes to AI.

5.3.7 Boundary Conditions

TE excludes consequence severity (CS), monitoring quality (CM), response execution (SOE), and resilience (RRAS).

5.3.8 Interpretation

A high TE score indicates greater modeled susceptibility to adversarial action under the available evidence. TE contributes directly to hazard pressure through Equation 20.

5.4 Security Operations Effectiveness (SOE)

5.4.1 Purpose and Philosophy

Security Operations Effectiveness (SOE) captures an organization's demonstrated ability to detect, respond to, remediate, and safely deploy changes in the presence of adversarial activity. SOE represents the operational execution dimension of cyber risk.

SOE is intentionally defined as an outcome-oriented capability rather than a measurement of specific tools, team structures, or processes. This ensures applicability across traditional IT operations, DevOps, SecOps, and future engineering paradigms.

5.4.2 Construct Definition

SOE captures the organization's ability to convert awareness into effective action. Where CM asks whether the organization can observe and verify security-relevant state, SOE asks whether the organization can act on that knowledge quickly, safely, and repeatably. It therefore represents the execution side of mitigation capacity: the operational pathway by which detected issues are triaged, contained, remediated, validated, and converted into durable improvement.

SOE captures four stable operational invariants:

1. **Detection Effectiveness**: the ability to identify meaningful security events, suspicious behavior, exposed weaknesses, and control failures quickly enough for action to matter. This includes timeliness of alerting, quality of triage, and reduction of blind or noisy detection pathways. Detection effectiveness differs from CM because it measures operational recognition and escalation, not merely the existence or freshness of telemetry.

2. **Response and Remediation Velocity**: the ability to contain incidents, close vulnerabilities, restore secure configuration, and complete corrective actions within timeframes appropriate to the hazard. This invariant includes mean time to detect, respond, remediate, and patch, but its conceptual focus is elapsed exposure time: how long a known or actionable problem remains unresolved.

3. **Change Safety and Reliability**: the ability to introduce defensive, corrective, and routine engineering changes without creating avoidable outages, regressions, misconfigurations, or new exposure. This invariant captures whether the organization can move quickly without increasing fragility through failed changes, rollbacks, emergency fixes, or uncontrolled exceptions.

4. **Operational Learning and Throughput**: the ability to sustain a reliable cadence of improvement, absorb lessons from incidents and near misses, and convert findings into repeatable engineering and security practices. This includes deployment regularity, preventive quality controls, post-incident learning, and security-gate effectiveness.

The construct is intentionally outcome-oriented. SOE does not require a particular SOC model, ticketing platform, CI/CD toolchain, or staffing structure. A centralized security operations center, a product-aligned SecOps model, and a highly automated cloud-native operation can all score well

if they demonstrate the same underlying capability: fast recognition, disciplined response, safe change, and continuous improvement. Conversely, strong tooling alone does not imply high SOE if issues remain untriaged, fixes stall, emergency changes frequently fail, or lessons do not alter future behavior.

Within the CRPI, SOE is the execution capability that is modulated by CM through Equation 24. This separation is deliberate. Visibility without execution does not remediate hazards, while execution with weak visibility can be misdirected or incomplete. The parameter m_0 allows an empirically governed baseline for response pathways that do not depend on continuous telemetry instead of forcing mitigation to zero whenever CM is zero.

5.4.3 Metric Form

SOE is modeled as

$$SOE(t) = \sum_{\ell=1}^{s} w_\ell^{SOE} \widetilde{SOE}_\ell(t), \qquad \sum_{\ell=1}^{s} w_\ell^{SOE} = 1, \tag{7}$$

where each $\widetilde{SOE}_\ell(t) \in [0, 1]$. Each SOE primitive should correspond to a demonstrated outcome, such as successful detection exercises, verified containment performance, safe changes, or closure of corrective actions. Composition weights express a documented construct judgment; they are not evidence of causality or universal efficacy. Shared event records between monitoring and response must be audited to avoid counting one control effect as independent evidence twice.

5.4.4 Measurement Interfaces and Representative Inputs

Representative sources include incident and case-management systems, vulnerability and patch-management systems, CI/CD and deployment logs, engineering quality artifacts, and supply-chain transparency artifacts. The recommended discrete inputs are:

Table 6: Recommended discrete inputs for Security Operations Effectiveness

Discrete input	Recommended evidence or input	Normalization	Cadence
Detection timeliness	MTTD or equivalent time from observable event to validated detection.	Lower time is better.	Event-driven
Response and containment time	Time from validated detection to containment or safe response.	Lower time is better.	Event-driven
Remediation and patch time	MTTP or equivalent time from actionable finding to verified remediation.	Lower time is better.	Event-driven
Change failure rate	Failed, rolled-back, emergency, or security-regressing changes relative to total changes.	Lower rate is better.	Per change/release
Mitigation throughput	Volume and timeliness of completed triage, containment, remediation, and validation work.	Higher verified throughput is better.	Daily/weekly

Discrete input	Recommended evidence or input	Normalization	Cadence
Preventive quality coverage	Fraction of changes and controls receiving required tests, reviews, and security gates.	Higher coverage is better.	Per change/release
Operational learning closure	Near-miss and post-incident findings converted into verified corrective actions.	Higher closure is better.	Weekly/event-driven
Supply-chain execution assurance	Timeliness and completeness of supplier, dependency, and build-pipeline corrective actions.	Higher assurance is better.	Weekly/change-driven

5.4.5 Boundary Conditions

SOE excludes threat environment (TE), telemetry quality (CM), impact severity (CS), and structural resilience or dependency trust (RRAS).

5.4.6 Interpretation

A high SOE score indicates an organization capable of rapidly detecting, responding to, and remediating security issues while sustaining safe and continuous change. SOE directly influences effective mitigation $M(t)$ and thereby affects both instantaneous residual posture and the dissipation of persistent exposure.

5.5 Resilience and Risk Assurance Score (RRAS)

5.5.1 Purpose and Philosophy

RRAS measures an organization's ability to withstand adverse events, limit cascading failure, and recover essential functions while maintaining assurance over critical dependencies. RRAS represents the structural survivability dimension of cyber risk.

RRAS is intentionally separated from day-to-day operational execution. While SOE measures how quickly teams detect and respond, RRAS measures whether the system can recover at all and whether its dependencies can be trusted during disruption.

5.5.2 Construct Definition

RRAS captures two interacting invariant constructs:

1. **Internal Resilience**: Structural properties enabling continued operation and recovery under failure.

2. **Dependency and Assurance Resilience**: The degree of trust, transparency, and control over dependencies whose failure could propagate risk.

5.5.3 Metric Form

The Resilience and Risk Assurance Score (RRAS) supplies one summary of recoverability, continuity, and dependency assurance. That summary supports reporting but does not determine which

specific consequences can be mitigated. The regularized geometric operator constructs the scalar report:

$$RRAS(t) = \mathcal{G}_{\varepsilon_{RRAS}}\left(\widetilde{Y}_1(t), \ldots, \widetilde{Y}_r(t); \mathbf{w}^{RRAS}\right), \qquad \sum_i w_i^{RRAS} = 1. \tag{8}$$

For the governing hazard equation, the same resilience primitives are projected onto each consequence domain using a governed domain-specific weight vector:

$$R_j(t) = \mathcal{G}_{\varepsilon_R}\left(\widetilde{Y}_1(t), \ldots, \widetilde{Y}_r(t); \mathbf{a}_j\right), \qquad a_{ji} \geq 0, \qquad \sum_i a_{ji} = 1. \tag{9}$$

The domain-specific projection R_j uses the same primitive resilience observations with weights appropriate to each kind of consequence. Restoration tests might matter strongly for availability while supplier attestations cannot alone establish recovery capability. The separate factor ρ_j controls the maximum mitigable fraction of that consequence, not the composition of the resilience evidence. The vectors $\mathbf{a}_j$ govern how the common resilience primitives are composed for each consequence domain; they are composition weights, not mitigability parameters. The resulting $R_j(t)$ is the domain-applicable resilience capability. By contrast, ρ_j is the maximum mitigability of domain j: it governs how much of that capability may reduce the domain's consequence in D_{eff}. This distinction prevents, for example, strong dependency assurance from fully compensating for weak recoverability in an availability consequence domain. The regularized operator retains a multiplicative penalty for weak components proportional to their governed construct weights, without asserting that a single zero-valued component makes all organizational resilience zero.

5.5.4 Measurement Interfaces and Representative Inputs

Representative sources include architecture and dependency inventories, recovery and continuity artifacts, supply-chain and dependency assurance artifacts, and governance records. The recommended discrete inputs are:

Table 7: Recommended discrete inputs for Resilience and Risk Assurance

Discrete input	Recommended evidence or input	Primary capability	Cadence
Recoverability capability	Restore tests, recovery exercises, RTO/RPO performance, and verified recovery dependencies.	Recovery	Event or scheduled
Redundancy and failover coverage	Independent capacity, failover paths, test results, and coverage of critical components.	Continuity	Daily or scheduled
Diversity and independence	Common-mode dependency concentration across providers, regions, platforms, and teams.	Fault tolerance	Change or review
Dependency transparency and provenance	Authoritative dependency inventory, lineage, ownership, component provenance, and change visibility.	Assurance	Change-driven

Discrete input	Recommended evidence or input	Primary capability	Cadence
Dependency trustworthiness	Supplier assurance, service commitments, control evidence, incident history, and validation status.	Assurance	Monthly/event-driven
Assured change integrity	Review, testing, approval, rollback, and post-change verification for critical changes.	Change resilience	Per change/release

5.5.5 Boundary Conditions

RRAS excludes incident response speed (SOE), exploitability signal (TE), impact severity (CS), and monitoring coverage (CM).

5.5.6 Interpretation

A high RRAS score indicates a system that is structurally resilient, recoverable, and supported by trustworthy and transparent dependencies. Within the governing model, the domain projection R_j modifies consequence through the consequence-specific mitigability factors ρ_j in Equation 19; it does not automatically erase all forms of impact.

6 Measurement Specification

This section defines how Tier 2 constructs are operationalized from telemetry, artifacts, and governance evidence. Its purpose is not to restate Tier 2 philosophy, but to specify the measurement layer that makes the CRPI implementable, testable, and governable in practice.

The measurement layer is intentionally separated from the governing equations. This separation is central to the CRPI's invariance claim: the equations remain fixed, while the evidence sources, normalization profiles, sampling cadences, and sub-metric instantiations may evolve under governance. In this way, the CRPI can accommodate new technologies, telemetry sources, or operating models without requiring a change to the mathematical structure of risk.

6.1 Common Measurement Conventions

Normalization. Each raw measure x is converted to a normalized score $\widetilde{x} \in [0, 1]$ using bounded mapping with explicit directionality:

$$\widetilde{x} = \text{clip}\left(\frac{x - L}{U - L}, 0, 1\right) \tag{10}$$

for higher-is-better quantities, and

$$\widetilde{x} = 1 - \text{clip}\left(\frac{x - L}{U - L}, 0, 1\right) \tag{11}$$

for lower-is-better quantities. Bounds must be versioned, reviewable, and associated with the calibration profile under which a score was produced.

In these equations, x is a raw observation with a declared unit, L and U are versioned lower and upper calibration anchors with $U > L$, and $\text{clip}(z, 0, 1)$ restricts the mapped value to the unit interval. For example, a measured containment time is a lower-is-better quantity: its normalized capability decreases as the measured time approaches the upper bound. A fraction such as 11 exposed systems out of 20 known internet-facing systems is already bounded, but its denominator and scope remain measurement assumptions. Normalization does not prove the measured population is complete.

Construct aggregation. Weighted sums are used where components are compensatory parts of a single construct. The regularized geometric operator in Equation 16 is reserved for constructs where weak components should materially reduce the aggregate in proportion to their governed weights without creating an unintended absorbing zero. The choice of aggregation operator is therefore part of construct semantics and must be documented rather than selected for numerical convenience. In particular, the operator retains a multiplicative penalty for weak components proportional to their governed construct weights; it does not guarantee that one weak component dominates the aggregate.

Smoothing. For volatile measures, an exponentially weighted moving average (EWMA) may be applied:

$$s_t = \alpha \widetilde{x}_t + (1 - \alpha) s_{t-1}, \qquad \alpha \in (0, 1]. \tag{12}$$

Smoothing is a measurement-estimation choice, not a change to construct semantics.

Missing Data. Missingness is treated explicitly rather than silently ignored. For any sub-metric $\widetilde{x}$, an availability indicator $a \in \{0, 1\}$ is maintained. If unavailable, the sub-metric is imputed using one of three governed policies:

1. *Last observation carried forward* (LOCF) for slow-moving quantities;
2. *Neutral imputation* where no directional assumption is justified;
3. *Risk-averse imputation* where lack of evidence plausibly implies lack of control, lack of assurance, or lack of visibility.

Sampling Frequency. Each sub-metric has an expected cadence. High-frequency measures are aggregated to the reporting interval using robust statistics (e.g., median, quantiles, bounded moving averages), while slow-moving artifacts are evaluated on change events or scheduled review cycles.

6.2 Coverage and Evidence Qualification

The CRPI should not be reported as a naked scalar without qualification of evidence quality. For each Tier 2 metric $X \in \{CM, CS, TE, SOE, RRAS\}$, define a coverage score $C_X(t) \in [0, 1]$ as the governed fraction of expected evidence that is observed and verifiable at time t. For each consequence domain $j \in \mathcal{D}$, also define $C_{D_j}(t)$ for evidence supporting $D_j(t)$ and $C_{R_j}(t)$ for evidence supporting the corresponding resilience projection $R_j(t)$. Report both

$$\bar{C}_{\text{CRPI}}(t) = \frac{1}{5} \sum_{X} C_X(t) \tag{13}$$

and

$$C_{\min}(t) = \min_X C_X(t). \tag{14}$$

The governing path has an additional hard evidence-completeness diagnostic:

$$C_{\text{gov}}(t) = \min\left(C_{TE}(t), C_{CM}(t), \min_{j\in\mathcal{D}} C_{D_j}(t), \min_{j\in\mathcal{D}} C_{R_j}(t), C_{SOE}(t)\right). \tag{15}$$

Tier 2 coverage remains useful for construct-level reporting, but C_{gov} prevents a scalar RRAS or CS report from hiding a poorly observed consequence or resilience domain. Coverage gates qualify the report; they do not multiply or otherwise alter the CRPI value. The minimum gate prevents strong coverage in four constructs from masking the complete absence of evidence in a fifth construct. Any qualitative label such as High, Medium, or Low evidence coverage must require both an aggregate-coverage threshold, a minimum-per-construct threshold, and a governing-path threshold, plus governed recency and source-integrity criteria. Thresholds belong to the calibration profile and are not universal constants.

Declared coverage versus scope completeness. The coverage fractions above are conditional on an expected evidence universe supplied by the assessor. If 90 monitored assets are the only 90 assets in an organization's inventory, declared coverage is 100%. If an independent discovery process subsequently finds 30 additional unmonitored assets, the same 90 observed assets constitute only 75% of the now-known population. Even that value remains conditional on the expanded discovery scope. Neither ratio proves that all shadow IT, unregistered identities, unmanaged workloads, or external dependencies have been found.

Two kinds of assessment must therefore be kept separate: evidence presence within a declared universe, and independent assurance about the completeness of that universe. Candidate assurance sources include cloud-resource control planes, network observations, identity directories, endpoint management, asset ownership records, and independent service/dependency discovery. Differences between sources require reconciliation, timestamped scope boundaries, and disclosure of potential duplicate or unmatched entities. No combination of these sources guarantees identification of every unknown asset.

The present reference implementation accepts caller-supplied coverage fractions and its provisional pilot computes completeness over declared fields. It does not authenticate all inventory denominators or independently discover hidden scope. Its calculated C_X values and coverage gates therefore *qualify evidence under declared scope*; they do not certify organization-wide completeness. A fully qualified enterprise interpretation should be withheld when the population boundary is materially disputed or not independently defensible. The underlying normalized CM coverage primitive can also be biased by an incomplete denominator, changing the modeled mitigation and CRPI themselves. Coverage is deliberately not multiplied into the governing risk equation; adding such a factor would require separate mathematical and empirical justification.

A prospective study should report known-asset coverage, unresolved discovery discrepancies, source-reconciliation agreement, sensitivity to plausible unobserved scope, and whether the stated reporting boundary is supported by independent evidence. Where such quantities cannot be bounded defensibly, an unqualified whole-organization score is not supported.

Coverage is not statistical confidence. Coverage describes evidence completeness. It must not be represented as a statistical confidence interval. Where sub-metrics have defensible uncertainty ranges or probability distributions, implementations should propagate those uncertainties through the CRPI equations, for example by Monte Carlo simulation, and report a distribution or quantile

interval for $CRPI(t)$. Such intervals characterize measurement/model uncertainty; they still do not convert CRPI into a probability of breach.

For temporal analyses, uncertainty propagation should evolve sampled trajectories through the persistence state $S(t)$; independently resampling each reporting interval would discard temporal dependence and is not equivalent to uncertainty propagation through the CRPI state model.

6.3 Measurement Governance and Internal Weights

The CRPI equations retain the same relationships between constructs even when organizations use different measurement sources and explicit calibration profiles. For example, different organizations may measure response effectiveness with different independent exercise and incident records. Both are potential SOE evidence if the scope, units, normalization, and measurement reliability are established.

Composition weights and calibration coefficients are therefore not universal constants. They must be versioned, justified by the meaning of the construct or observed independent evidence, sensitivity tested, and reviewed before deployment. They are not discretionary controls used to make the headline CRPI value appear desirable. Equal scores under different calibration profiles need not imply equivalent operational conditions.

6.4 Minimum Operational Inputs and Partial Evidence

The algebraic instantaneous residual needs only governed TE, D_{eff}, and M, since $R_{\text{inst}} = TE\, D_{\text{eff}}(1 - M)$. These three numbers do not, however, provide operational traceability to measured exposures, consequence domains, resilience, or control execution. A decomposed evaluation requires TE, each D_j, each R_j for which $\rho_j > 0$, CM, and SOE, plus governed weights, ρ_j, and m_0. Under the illustrative three-domain profile, $\rho_{\text{confidentiality}} = 0$, so eight normalized construct/domain scores suffice for the instantaneous residual. Nine would be required if all three ρ_j were nonzero. These are already normalized and governed scores, not eight raw counters.

The executable reference runner obtains these constructs from 15 normalized primitive fields: two TE, three consequence, four CM, four SOE, and two resilience inputs. Domain resilience projections R_j use the two resilience primitives and domain-specific weights; scalar RRAS is reported separately, not multiplied into the hazard calculation. A full temporal CRPI additionally requires prior state (or an explicitly declared cold-start assumption), elapsed time, and the two rate parameters in matching time units. Table 8 records the provisional 15-primitive measurement profile used in the synthetic raw-evidence pilot. This mapping is specified here so that the paper remains independently interpretable. The associated public repository contains an executable illustration, but neither the candidate normalizers nor their source denominators constitute empirically calibrated organization-wide measurements.

Table 8: Provisional 15-input operational measurement contract; mappings and bounds require study validation

Primitive	Candidate evidence and raw observation	Provisional normalization / primary limitation
TE_{external}	Validated actionable external exposures among declared internet-facing systems	Adverse fraction; independent asset census and relevance required.

Primitive	Candidate evidence and raw observation	Provisional normalization / primary limitation
TE_{internal}	Independently assessed unsafe privileged or reachable paths	Adverse fraction; assessed-only denominator is selection-sensitive.
CS_{conf}	Independently reviewed disclosure/inference consequence class	Ordinal 0–4 divided by four; spacing and anchors provisional.
$CS_{\text{integrity}}$	Mission/decision integrity consequence class	Ordinal 0–4 divided by four; scope and impact horizon governed.
$CS_{\text{availability}}$	Continuity/interruption consequence class	Ordinal 0–4 divided by four; impact assumptions reviewed.
CM_{coverage}	Required controls with verified monitoring / declared required controls	Capability fraction; unknown control universe can inflate result.
$CM_{\text{freshness}}$	P90 evidence ingestion/validation delay (hours)	Inverse bounded 0–24 h illustrative map; censoring/sample size matter.
$CM_{\text{integrity}}$	Required streams passing provenance/integrity checks	Capability fraction; attestations need independent verification.
$CM_{\text{verification}}$	Critical monitored controls with independently verified true state	Capability fraction; avoid reusing SOE outcome evidence.
$SOE_{\text{detection}}$	Controlled detection exercises successfully triaged / eligible attempts	Capability fraction; exercise difficulty and sampling govern validity.
SOE_{velocity}	P90 hours from eligible trigger to verified containment	Inverse bounded 0–72 h illustrative map; censoring and small samples matter.
SOE_{change}	Safe verified changes without rollback/security regression / eligible changes	Capability fraction; reconcile changes and consequences.
SOE_{learning}	Due corrective actions independently closed / all due actions	Capability fraction; no due actions is missing, not 100%.
$RRAS_{\text{internal}}$	Critical recovery exercises meeting documented objectives / eligible exercises	Capability fraction; attested plan is not tested recovery.
$RRAS_{\text{dependency}}$	Critical dependencies with independently reviewed assurance / declared dependencies	Capability fraction; dependency census and assurance evidence need audit.

The source names shown in this table are scientific shorthand; canonical executable field identifiers are `TE_external`, `TE_internal`, `CS_confidentiality`, `CS_integrity`, `CS_availability`, `CM_coverage`, `CM_freshness`, `CM_integrity`, `CM_verification`, `SOE_detection`, `SOE_velocity`, `SOE_change_safety`, `SOE_learning`, `RRAS_internal`, and `RRAS_dependency`. The illustrative time caps, ordinal mappings, fractions, and source-identity assertions must be qualified independently before organizational use.

Evidence incompleteness must be distinguished from low observed security scores. The present executable policy rejects missing required normalized inputs. Last-observation, neutral, and risk-averse imputation discussed above are possible *governed alternative* policies, not actions the current

runner performs silently. Partial outputs and mathematically admissible possible-value intervals can be reported with explicit assumptions, but should not be relabeled as a fully measured CRPI or as statistical confidence intervals. Every output also needs scope, profile version, component lineage, evidence qualification, and a cold-start history indicator where applicable.

6.5 Tier 2 Measurement Summary

Table 9 provides a compact summary of the Tier 2 metrics and their measurement roles.

Table 9: Tier 2 metric summary

Tier 2	Risk Role	Example Inputs	Aggregation	Temporal Behavior
CM	Ability to know	telemetry coverage, freshness, integrity, drift detection	weighted sum	optional smoothing
CS	Consequence of failure	sensitivity, criticality, derived sensitivity, availability impact	weighted sum	slow-moving, change-driven
TE	Susceptibility to attack	attack surface, exploitability, identity exposure, intel relevance	regularized geometric + EWMA	moderately volatile
SOE	Ability to act	MTTD, MTTR, MTTP, change safety, throughput, quality checks	weighted sum	event-driven, optional smoothing
RRAS	Ability to survive	recoverability, redundancy, diversity, provenance, trustworthiness	scalar regularized geometric plus domain projections	slow-moving, structural updates

7 Temporal Behavior and Calibration

Cyber risk is non-stationary. Changes in adversary behavior, technology, organizational structure, and defensive capability change posture continuously rather than only at assessment boundaries. Time is therefore a first-class component of the CRPI.

Temporal behavior appears at two levels. First, volatile Tier 2 evidence may be smoothed using governed estimators such as EWMA. Second, the top-level persistence state $S(t)$ uses the bounded stock-flow model in Equation 28. The rate parameters β and κ have inverse-time units and the exact interval update explicitly uses Δt; this prevents the meaning of persistence from silently changing when an implementation moves from daily to hourly or other reporting cadences.

Calibration is required for longitudinal comparability, but recalibration must not be silent. Normalization bounds, α, ε_X, consequence-composition weights v_j, resilience-mitigability factors ρ_j, m_0, β, and κ are all versioned under a named calibration profile.

Each calibration profile should explicitly identify the model/version identifier, declared time unit, finite domain set $\mathcal{D}$, consequence weights v_j, resilience primitive definitions, domain-projection weights $\mathbf{a}_j$, mitigability factors ρ_j, normalization bounds, smoothing and regularization parameters, evidence thresholds, missing-data policy, and temporal-initialization policy. Without the profile identifier, longitudinal or cross-organizational comparisons are not interpretable as comparisons under the same model instantiation.

7.1 Calibration Governance

At a minimum:

- calibration profiles must be versioned and identify their time unit;
- historical scores must remain associated with the profile under which they were produced;
- recalibration triggers must be explicit, such as sustained distribution shift, material scope change, major architecture change, instrumentation change, or evidence of validation breakdown;
- composition parameters must be justified by construct semantics or evidence, not by a desire to move the headline score;
- parameter changes must be sensitivity tested against historical and synthetic scenarios before adoption; and
- when profiles change materially, organizations should either preserve the old series or recompute a clearly labeled comparable historical series.

Table 10: Implementation-oriented calibration guidance

Parameter	Initial analysis range/profile	Governance interpretation
α_{TE}	0.40–0.70 sensitivity range	Higher values react faster; lower values damp noisy threat feeds.
α for CM/SOE	0.20–0.50 sensitivity range	Moderate smoothing may reduce reaction to short operational fluctuations.
ε_X	0.01–0.10 sensitivity range	Keep small; select per construct and verify that zero-safety does not over-soften severe weakness.
m_0	No universal default	Estimate the portion of operational effectiveness demonstrably independent of continuous telemetry; sensitivity test broadly.
ρ_j, $\mathbf{a}_j$	Domain-specific in $[0, 1]$	ρ_j encodes demonstrated consequence mitigability; $\mathbf{a}_j$ governs the resilience-primitive projection for each domain, not risk appetite.
β, κ	Positive rates in declared time units	Fit or backtest against observed accumulation and remediation dynamics; do not copy values across cadences without unit conversion.
Normalization bounds	Annual and event-driven review	Reassess after material scope, telemetry, architecture, or adversary changes.
Evidence gates	Profile-defined $\bar{C}_{\mathrm{CRPI}}$, $C_{\min}$, and C_{gov} thresholds	Aggregate, minimum, and governing-path coverage must pass before a high-coverage label is allowed.
AI-era review	Quarterly or after material threat shift	Shorten review windows when evidence shows discovery-to-exploitation dynamics have changed.

8 Governing Model

The governing model is now specified after the construct, measurement, and temporal definitions so that each variable has an operational interpretation. The governing model is intentionally compact. Its equation form is fixed; calibration parameters and measurement instantiations are governed configuration. The model does not claim that these equations are universal laws of cyber-risk behavior. They are structurally motivated relationships chosen to enforce explicit semantics, boundedness, decomposability, and testable temporal behavior.

This section is the calculation specification. At each reporting time, normalized evidence first instantiates the Tier 2 constructs CM, CS, TE, SOE, and RRAS. The consequence domains and resilience projections are then kept separate long enough to calculate D_{eff} and H. CM and SOE determine effective mitigation M; R_{inst} and the prior stock determine the updated persistent state S; and the final posture output is the envelope $CRPI = \max(R_{\text{inst}}, S)$. This sequence is identical at every scope of analysis. In this paper, "objective" means deterministic, auditable, and reproducible under a declared evidence and calibration profile; it does not mean assumption-free or free of measurement judgment.

The calculation order is: normalize scoped evidence, construct the five Tier 2 dimensions, retain domain-specific consequences and resilience, calculate hazard pressure and effective mitigation, calculate instantaneous residual posture, update persistent unresolved exposure, and report the larger of the two residual channels. The equations below make the relationships inspectable, but neither their functional forms nor their illustrative parameters establish empirical measurement validity.

8.1 Zero-Safe Regularized Geometric Aggregation

For constructs whose components should interact multiplicatively, CRPI uses a regularized weighted geometric operator rather than an ordinary geometric mean. For non-negative normalized weights $w_i \geq 0$ satisfying $\sum_i w_i = 1$,

$$\mathcal{G}_\varepsilon(\mathbf{x}, \mathbf{w}) = \frac{\prod_{i=1}^{n} [\varepsilon + (1-\varepsilon)x_i]^{w_i} - \varepsilon}{1-\varepsilon}, \qquad x_i \in [0,1], \quad \varepsilon \in (0,1). \tag{16}$$

The operator maps to $[0,1]$, returns zero when all positively weighted inputs are zero, returns one when all are one, and does not allow a single zero-valued component to collapse an otherwise non-zero construct. It is non-decreasing in every component; strict directional sensitivity is asserted only for components with $w_i > 0$. Small ε preserves the intended geometric penalty for weak components. ε is therefore a versioned construct-calibration parameter rather than a universal constant.

8.1.1 Calculation walkthrough

An operational calculation begins with source records rather than preselected scores. For a synthetic illustration, 11 actionable exposed systems out of 20 in-scope internet-facing systems yield a normalized external exposure input of $11/20 = 0.55$. An independent path assessment finding 13 unsafe privileged paths out of 20 assessed paths yields a normalized internal input of $13/20 = 0.65$. The choice of denominator and eligible exposure definition can substantially alter both ratios, particularly when discovery is incomplete. These ratios are illustrative proxies, not attack probabilities.

The two bounded values can then be composed with equal weights and a regularizer that reduces numerical collapse from one valid zero while retaining a penalty for weak components. For the normalized values above, with $\mathbf{x} = (0.55, 0.65)$, $\mathbf{w} = (0.5, 0.5)$, and $\varepsilon = 0.05$,

$$\mathcal{G}_{0.05}(\mathbf{x}, \mathbf{w}) = \frac{\sqrt{(0.05 + 0.95 \cdot 0.55)(0.05 + 0.95 \cdot 0.65)} - 0.05}{0.95} \approx 0.598. \tag{17}$$

This is the illustrative TE calculation used by the reference profile, rounded to $TE \approx 0.60$. The same operator is used for scalar RRAS and for each R_j when the construct semantics call for a multiplicative penalty. A zero-weight component is ignored by the operator; a zero-valued component with positive weight contributes a penalty without becoming an unintended absorbing zero.

A consequence score quantifies the severity of harm if compromise occurs, independently of whether compromise is likely. Hazard pressure is the model's bounded combination of threat exposure and remaining consequence before current operational mitigation is applied. Neither quantity is a calibrated breach probability.

8.2 Consequence and Hazard Pressure

The reporting aggregate for consequence is

$$CS(t) = \sum_{j \in \mathcal{D}} v_j D_j(t), \qquad v_j \geq 0, \qquad \sum_{j \in \mathcal{D}} v_j = 1. \tag{18}$$

The governing hazard calculation retains the consequence domains long enough to avoid assuming that resilience mitigates all forms of harm equally:

$$D_{\text{eff}}(t) = \sum_{j \in \mathcal{D}} v_j D_j(t) \left[1 - \rho_j R_j(t)\right], \qquad \rho_j \in [0, 1]. \tag{19}$$

The resilience-adjusted consequence D_{eff} is assembled from three kinds of input. The domain-specific severity D_j describes potential harm; the non-negative weight v_j determines that domain's contribution; and $1 - \rho_j R_j$ retains the portion of harm not mitigated by applicable structural resilience. Here R_j measures relevant resilience capability while ρ_j is a separate assumption about the maximum mitigable fraction of that consequence. These choices are governed, inspectable hypotheses, not universal empirical constants. A baseline profile may use confidentiality/sensitivity, integrity/mission, and availability/continuity domains, while other environments may define a different finite governed domain set $\mathcal{D}$. The ρ_j values encode the extent to which structural resilience can actually reduce the realized consequence in each domain, while $R_j(t)$ projects the heterogeneous resilience primitives onto the same consequence domain. For example, recoverability may strongly influence availability consequence while dependency assurance may be more relevant to confidentiality consequence; neither projection is assumed to be universally interchangeable.

Hazard pressure is then

$$H(t) = TE(t)\, D_{\text{eff}}(t). \tag{20}$$

Multiplying the bounded exposure and adjusted consequence produces a bounded hazard score and preserves the expected direction of each input. The domain adjustment avoids treating structural resilience as capable of erasing every consequence. For example, restoring service availability after disruption does not undo an already completed disclosure of confidential records. A resilience projection must therefore be relevant to the specific form of harm before it reduces consequence. This mathematical choice remains subject to empirical assessment.

8.2.1 Calculation walkthrough

Calculate the reporting consequence score first, then calculate the resilience-adjusted consequence used by the hazard path. For an illustrative three-domain row with equal consequence weights,

$$CS = \frac{0.90 + 0.80 + 0.70}{3} = 0.800, \tag{21}$$

$$D_{\text{eff}} = \frac{1}{3} \sum_{j \in \{C,I,A\}} D_j(1 - \rho_j R_j) \approx 0.582, \tag{22}$$

$$H = TE\, D_{\text{eff}} = 0.598 \cdot 0.582 \approx 0.348. \tag{23}$$

Here C, I, and A denote confidentiality, integrity, and availability consequence domains; the illustrative resilience projections are approximately $(R_C, R_I, R_A) = (0.655, 0.675, 0.695)$ and the mitigability factors are $(\rho_C, \rho_I, \rho_A) = (0, 0.4, 0.9)$. The zero confidentiality factor leaves that consequence unchanged by resilience, while the integrity and availability terms are reduced according to their governed mitigability. CS is therefore a useful report, but D_{eff} is the value that enters H.

8.3 Observability-Modulated Mitigation

Continuous Monitoring (CM) describes the credibility and continuity of observability, while Security Operations Effectiveness (SOE) measures demonstrated capacity to detect, respond, remediate, and safely deploy security-relevant changes. Observing a vulnerability does not fix it. Likewise, an organization with effective response procedures may still act too slowly or incompletely when it cannot reliably observe events. The model treats visibility as a modifier of operational execution rather than as an independent reduction in consequence:

$$M(t) = SOE(t)\,[m_0 + (1 - m_0)CM(t)]\,, \qquad m_0 \in [0, 1]. \tag{24}$$

The parameter m_0 represents the governed fraction of operational effectiveness that can exist without continuous telemetry, such as manual reporting, scheduled testing, or other non-continuous pathways. With all inputs in $[0, 1]$, $M(t)$ remains in $[0, 1]$.

The monitoring-independent fraction m_0 allows some operational effectiveness without continuous telemetry, such as scheduled assessments, manual escalation, and out-of-band reports. When $SOE = 0$, mitigation is zero regardless of monitoring. When $CM = 1$, mitigation equals SOE; when $CM = 0$, it is limited to $SOE\, m_0$. The multiplication encodes a complementary relationship between observability and action. Its precise form and coefficient are modeling assumptions that should be compared with simpler interaction forms in subsequent validation. Instantaneous residual posture is

$$R_{\text{inst}}(t) = H(t)\,[1 - M(t)]\,. \tag{25}$$

8.3.1 Calculation walkthrough

Using the preceding illustrative hazard value, suppose $CM = 0.725$, $SOE = 0.600$, and $m_0 = 0.25$. Then

$$M = 0.600\,[0.25 + 0.75(0.725)] \approx 0.476, \tag{26}$$

$$R_{\text{inst}} = 0.348(1 - 0.476) \approx 0.182. \tag{27}$$

The calculation makes the metric assignments explicit: TE and consequence determine hazard pressure, while CM and SOE determine how much of that hazard can be mitigated in the current interval. CM does not independently subtract consequence, and SOE does not replace the threat or consequence measurements.

Current residual posture R_{inst} cannot describe unresolved effects carried forward from earlier periods. The persistent unresolved exposure state $S(t)$ is therefore a normalized stock-like quantity bounded between zero and one. It is not a literal count of unresolved vulnerabilities, a measured incident backlog, or an attack probability. Residual exposure contributes inflow, while effective mitigation can retire part of the existing state. The model treats both flows as functions of observed state and governed rates, not as empirically established universal laws.

8.4 Persistent Unresolved Exposure

Persistent exposure is modeled as a bounded stock with unresolved instantaneous posture as inflow and effective mitigation as outflow:

$$\frac{dS}{dt} = \beta R_{\text{inst}}(t)\,[1 - S(t)] - \kappa M(t) S(t), \qquad \beta, \kappa \geq 0. \tag{28}$$

The inflow rate is $\beta R_{\text{inst}}(1 - S)$: the coefficient β sets how strongly current residual exposure contributes, while $(1-S)$ slows accumulation as the stock approaches its upper bound. The outflow rate is κMS: existing exposure is retired in proportion to the stock and effective mitigation, using the separate rate κ. Both rates have inverse-time units and should not be interpreted as calibrated probabilities. The $(1-S)$ term bounds accumulation as the stock approaches one. Under piecewise constant inputs over an interval Δt, define

$$q(t) = \beta R_{\text{inst}}(t) + \kappa M(t). \tag{29}$$

For $q(t) > 0$, the exact interval update is

$$S(t + \Delta t) = S^*(t) + [S(t) - S^*(t)]\, e^{-q(t)\Delta t}, \tag{30}$$

where

$$S^*(t) = \frac{\beta R_{\text{inst}}(t)}{q(t)}. \tag{31}$$

The exact solution assumes that residual exposure and mitigation are constant throughout each interval. The exponential term determines how quickly the starting stock approaches the interval-specific equilibrium. A sequence of changing daily inputs therefore cannot generally be replaced by their weekly average without losing event order. The interval Δt must be expressed in the same time unit as the rate parameters. If $q(t) = 0$, then $S(t + \Delta t) = S(t)$. This formulation preserves $S \in [0, 1]$, accumulates unresolved exposure when instantaneous residual posture is non-zero, dissipates accumulated exposure when mitigation is effective, and makes sampling cadence explicit through Δt. When historical state cannot be reconstructed, $S(t_0) = R_{\text{inst}}(t_0)$ provides a bounded state-neutral initialization. It makes no assertion about unresolved exposure preceding t_0. Outputs should therefore carry reduced temporal-history confidence until sufficient observation history accumulates or prior state is reconstructed.

Effective mitigation affects persistence through two distinct pathways. First, it reduces the rate at which current hazard becomes unresolved residual exposure through R_{inst}. Second, it represents the organization's capacity to remediate previously accumulated unresolved exposure through the $\kappa M(t) S(t)$ outflow term. These effects are intentionally separated as prevention or containment of new residual exposure and retirement of existing exposure stock.

8.4.1 Calculation walkthrough

The stock update uses the current interval's R_{inst} and M together with the prior stock. Continuing the illustrative calculation with $S(t) = 0.200$, $\beta = 0.30$, $\kappa = 0.80$, and $\Delta t = 1$,

$$q = 0.30(0.182) + 0.80(0.476) \approx 0.435, \tag{32}$$

$$S^* = \frac{0.30(0.182)}{0.435} \approx 0.125, \tag{33}$$

$$S(t+1) = 0.125 + (0.200 - 0.125)e^{-0.435} \approx 0.174. \tag{34}$$

The stock rises when unresolved posture supplies more inflow than mitigation removes, and falls when effective mitigation dominates. Because the update is exact for the piecewise-constant interval assumption, changing the reporting cadence requires changing Δt in the same declared time unit rather than changing the governing equation.

The final scalar reports the more adverse of the current residual and persistent-state channels. This is a deliberately conservative decision rule, not an additive risk total or a mechanistic model of incidents. Both channel values and the controlling branch must accompany the index so that the reason for a change remains interpretable.

8.5 Posture Output Envelope

The top-level CRPI value is the conservative envelope

$$CRPI(t) = \max\left(R_{\text{inst}}(t), S(t)\right). \tag{35}$$

The maximum is an envelope operator, not a mechanistic description of incident dynamics. It prevents recent operational improvement from masking material persistent exposure while keeping the result bounded and directly decomposable.

8.5.1 Calculation walkthrough

For the illustrative values above, $R_{\text{inst}} \approx 0.182$ and $S(t+1) \approx 0.174$, so the envelope is $CRPI(t+1) = 0.182$. If the persistent stock were 0.300 instead, the same current condition would produce $CRPI = 0.300$ and make the unresolved history the controlling explanation. The maximum is therefore not an executive-only display choice: it is the defined objective output of the state model, with executive reporting as one consumer of that output.

8.6 Scope Levels, Roll-Up, and Calibration Boundary

The governing equations do not change when the unit of analysis changes. A component, sub-system, system, network, organizational domain, or enterprise is treated as a scoped entity u with its own evidence boundary, consequence domains, resilience mapping, and persistent state. The same sequence is evaluated for that entity: construct its bounded Tier 2 values, calculate its domain-specific D_{eff}, then H, M, R_{inst}, S, and $CRPI$. The scope changes the evidence and the meaning of the result, not the equation form.

For a scoped entity u, let ℓ denote its analytical level: component, sub-system, system, network, domain, or enterprise. The final output is

$$CRPI_{u,\ell}(t) = \max(R_{\text{inst},u,\ell}(t), S_{u,\ell}(t)). \tag{36}$$

The component equations for TE, CS, SOE, $RRAS$, D_{eff}, H, M, and S retain their Section 8 form; each input is simply indexed by (u, ℓ) and must be calculated from that scope's evidence boundary. This makes the distinction between a directly measured parent risk value and a separately governed aggregation of child values explicit.

Table 11: CRPI application across analytical levels

Scope	Unit of analysis	Recommended evidence boundary and interpretation
Component	Asset, service, identity, control point, or data store	Direct telemetry, configuration state, local exposure, component consequence, and component recovery evidence.
Sub-system	Application, cluster, control plane, or service group	Component interactions, shared controls, local dependencies, subsystem mission impact, and response ownership.
System	Product, workload, plant, or business service	System attack surface, system-level consequence domains, end-to-end mitigation performance, and recoverability evidence.
Network	Segment, trust zone, connectivity plane, or interconnection	Reachability, trust boundaries, traffic and control evidence, lateral-exposure conditions, and cross-system dependency effects.
Domain	Mission, business, data, or operational domain	Domain portfolio consequence, shared services, domain dependencies, and the domain-specific resilience mappings relevant to that scope.
Enterprise	Organization or governed portfolio	Enterprise-wide evidence, shared infrastructure, cross-domain consequences, centralized and federated response, and enterprise resilience.

The scope-level output is objective in the operational sense: the same declared inputs, normalization rules, weights, and time history reproduce the same result and allow the result to be decomposed back to evidence. It is not assumption-free. A parent score should be calculated from parent-scope evidence whenever possible; averaging child CRPI values can hide concentration, shared dependencies, or a high-consequence outlier. If only child scores are available, publish any roll-up as a separately named and governed aggregation rather than silently presenting it as the parent CRPI.

The term "domain" in this scope hierarchy is distinct from a consequence domain $j \in \mathcal{D}$. A single enterprise domain may contain confidentiality, integrity, and availability consequence domains, while a single system may contribute to several organizational domains.

8.6.1 Calibration boundary

The governing equation form is stable, but the model is not parameter-free. The parameters ε_X, v_j, ρ_j, $\mathbf{a}_j$, m_0, β, κ, α, and normalization bounds are explicit calibration or construct-composition parameters. They must be versioned, justified by construct semantics or observed data, sensitivity

tested, and prohibited from ad hoc tuning solely to make the posture value appear desirable. Cross-level and cross-organization comparisons require the profile identifier and a documented scope definition; otherwise equal scores need not mean comparable things.

9 Interpreting CRPI Outputs

The top-level $CRPI(t)$ value is a bounded operational posture index in $[0, 1]$, where values closer to zero indicate lower modeled residual cyber-risk posture and values closer to one indicate higher modeled residual posture under a defined calibration profile. It is not a direct probability of breach or monetary-loss estimate.

The value should be read together with its decomposition:

- $H(t)$ answers: *How much hazard pressure exists before current operational mitigation?* It rises as threat exposure or resilience-adjusted consequence rises.
- $M(t)$ answers: *How much effective mitigation can the organization apply?* It reflects operational execution as modulated by the quality of continuous observability.
- $R_{\text{inst}}(t)$ answers: *What residual posture is implied by current conditions now?*
- $S(t)$ answers: *How much unresolved residual exposure remains from current and recent conditions?*
- $CRPI(t)$ reports the conservative residual-risk envelope used by operational, engineering, governance, and executive consumers.

Two organizations can have the same $CRPI(t)$ for materially different reasons. One may have high current threat pressure but low accumulated exposure; another may have moderate current hazard but substantial persistent stock. Management action therefore requires the decomposition, Tier 2 values, evidence coverage, and uncertainty qualification rather than the scalar alone.

Table 12: Score direction and operational interpretation

Quantity	Higher Means	Interpretation
$CM(t)$	Better	More complete, current, and trustworthy visibility.
$SOE(t)$	Better	Stronger ability to detect, respond, remediate, and change safely.
$RRAS(t)$	Better	Stronger scalar structural survivability and assurance summary.
$R_j(t)$	Better	Stronger resilience projection for consequence domain j.
$CS(t)$	More severe	Greater aggregate consequence if the system or data is compromised.
$TE(t)$	More exposed	Greater adversary pressure, exploitability, or attackability.
$D_{\text{eff}}(t)$	More severe	Greater consequence remaining after consequence-specific resilience adjustment.
$H(t)$	More hazardous	More inherent hazard before current operational mitigation.

Quantity	Higher Means	Interpretation
$M(t)$	Better	Greater effective mitigation capacity.
$R_{\text{inst}}(t)$	More residual risk	Worse current residual posture.
$S(t)$	More persistent	More unresolved exposure carried forward.
$CRPI(t)$	More residual risk	Higher modeled operational cyber-risk posture.

10 Worked Example

This section exercises the intended behavior of the revised governing equations under a small synthetic sequence. It is an illustrative calculation, not empirical validation.

Assume five equal intervals with $\Delta t = 1$. To keep the table compact, assume the consequence composition remains stable and, for this compact special case, each domain-applicable resilience projection equals the scalar report $R_j(t) = RRAS(t)$. The consequence-weighted resilience mitigability is $\bar{\rho} = 0.60$, making the domain equation reduce to

$$D_{\text{eff}}(t) = CS(t)\,[1 - 0.60\,RRAS(t)]\,. \tag{37}$$

Let

$$m_0 = 0.25, \qquad \beta = 0.30, \qquad \kappa = 0.80. \tag{38}$$

These are illustration parameters, not recommended universal defaults.

At t_0 let

$$TE = 0.60, \quad CS = 0.80, \quad RRAS = 0.40, \quad CM = 0.50, \quad SOE = 0.40. \tag{39}$$

Then

$$D_{\text{eff}}(0) = 0.80(1 - 0.60 \cdot 0.40) = 0.608, \tag{40}$$

$$H(0) = 0.60 \cdot 0.608 = 0.365, \tag{41}$$

$$M(0) = 0.40[0.25 + 0.75(0.50)] = 0.250, \tag{42}$$

$$R_{\text{inst}}(0) = 0.365(1 - 0.250) = 0.274. \tag{43}$$

Because no prior state is reconstructed in this illustrative sequence, use the bounded state-neutral initialization $S(0) = R_{\text{inst}}(0) = 0.274$, giving $CRPI(0) = 0.274$. This initialization makes no assertion about unresolved exposure before t_0; the illustrative outputs therefore carry reduced temporal-history confidence until sufficient observation history accumulates or prior state is reconstructed.

At t_1, threat exposure rises to $TE = 0.80$ while the other Tier 2 scores remain unchanged. Instantaneous residual posture rises to 0.365. The persistence stock also rises, but more gradually, to 0.295, so the current condition controls the posture envelope.

At t_2, monitoring and operational execution improve to $CM = 0.75$ and $SOE = 0.70$. Effective mitigation rises to $M = 0.569$, reducing instantaneous residual posture to 0.189. Persistent stock remains higher at 0.221, so the posture envelope does not immediately forget the recent period of elevated unresolved exposure.

By t_3 and t_4, threat exposure moderates and resilience improves. Instantaneous posture continues to fall while the persistent channel dissipates more gradually.

10.1 Expanded Time-Series View

The two tables separate the measured or assumed Tier 2 inputs from the derived state and output. Each input is a bounded illustrative score; none is a direct probability. The signed rows below each new time step show changes from the preceding step, so that a reader can identify which input moved before interpreting the response in hazard, mitigation, or persistence.

Table 13: Illustrative Tier 2 input values and changes relative to the preceding observation

Time	TE	CS	$RRAS$	CM	SOE
t_0	0.600	0.800	0.400	0.500	0.400
t_1	0.800	0.800	0.400	0.500	0.400
Δ_1	+0.200	0.000	0.000	0.000	0.000
t_2	0.750	0.800	0.450	0.750	0.700
Δ_2	−0.050	0.000	+0.050	+0.250	+0.300
t_3	0.650	0.780	0.550	0.820	0.780
Δ_3	−0.100	−0.020	+0.100	+0.070	+0.080
t_4	0.550	0.750	0.650	0.880	0.840
Δ_4	−0.100	−0.030	+0.100	+0.060	+0.060

Table 14: Derived consequence, hazard, mitigation, persistent state and CRPI; all values rounded to three decimals

Time	D_{eff}	H	M	R_{inst}	S	$CRPI$	$\Delta CRPI$
t_0	0.608	0.365	0.250	0.274	0.274	0.274	–
t_1	0.608	0.486	0.250	0.365	0.295	0.365	+0.091
t_2	0.584	0.438	0.569	0.189	0.221	0.221	−0.144
t_3	0.523	0.340	0.675	0.111	0.150	0.150	−0.071
t_4	0.458	0.252	0.764	0.059	0.093	0.093	−0.057

Here Δ_k denotes the change from t_{k-1} to t_k in the input table, not the elapsed duration Δt used by the stock-flow equation. Changes are arithmetic differences between displayed rounded values, not percentages. A positive increase in TE indicates more modeled exposure, while a positive change in CM or SOE indicates stronger modeled defensive capability. This is why the input deltas are not assigned a universal favorable/unfavorable color. At t_2, current residual posture falls below persistent exposure, and the persistent state becomes the controlling branch of the reported index.

10.2 Interpretation of Model Behavior

Four behaviors are visible. First, higher TE immediately increases H and R_{inst}. Second, stronger CM improves mitigation only through its effect on operational execution; CM alone is not modeled as remediation. Third, stronger domain-applicable resilience projections R_j reduce only the consequence that the governed ρ_j terms identify as mitigable. Fourth, $S(t)$ retains recent unresolved exposure after instantaneous posture improves and then dissipates under stronger mitigation.

The stock-flow boundary cases are also concrete. If $R_{\text{inst}} > 0$ and $M = 0$, S monotonically approaches one. If $R_{\text{inst}} = 0$ and $M > 0$, S decays toward zero. If both are zero, the stock

remains unchanged. These are the intended semantics of unresolved exposure accumulation and remediation.

10.3 Figures

Figure 2 shows distinct Tier 2 input trajectories using both color and marker/line style. Figure 3 shows hazard, effective mitigation, and persistent stock. Figure 4 shows instantaneous residual posture, persistent exposure, and the posture envelope.

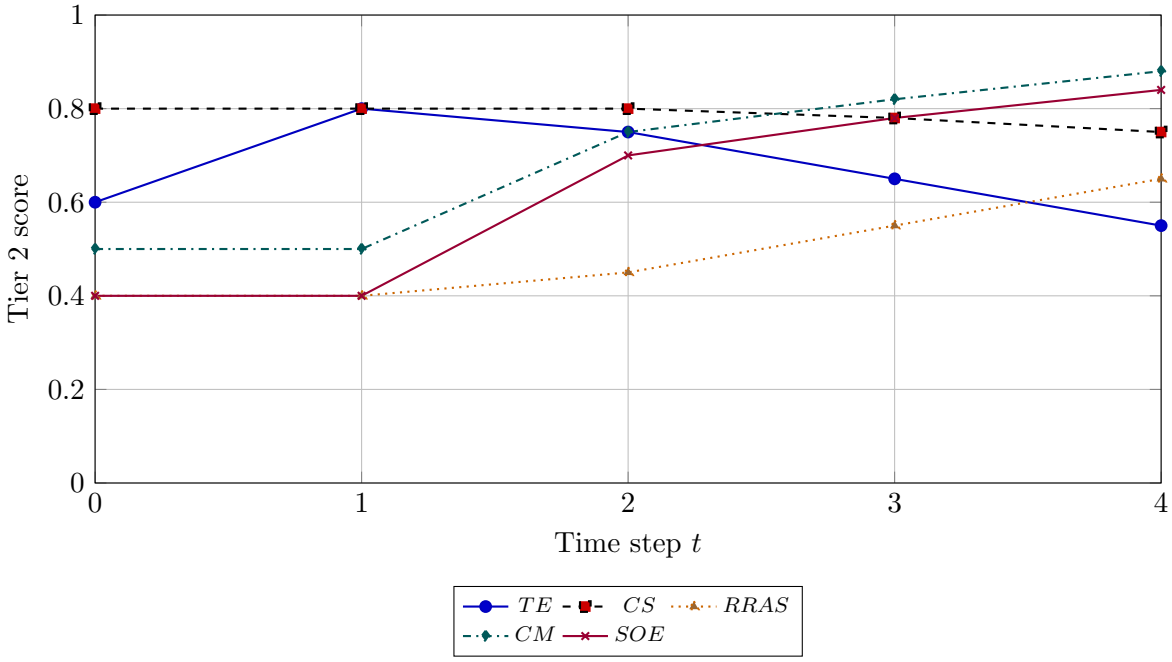


Figure 2: Five normalized Tier 2 trajectories in the synthetic worked example. Independent colors, line styles, and marker shapes distinguish the series even when reproduced in grayscale.

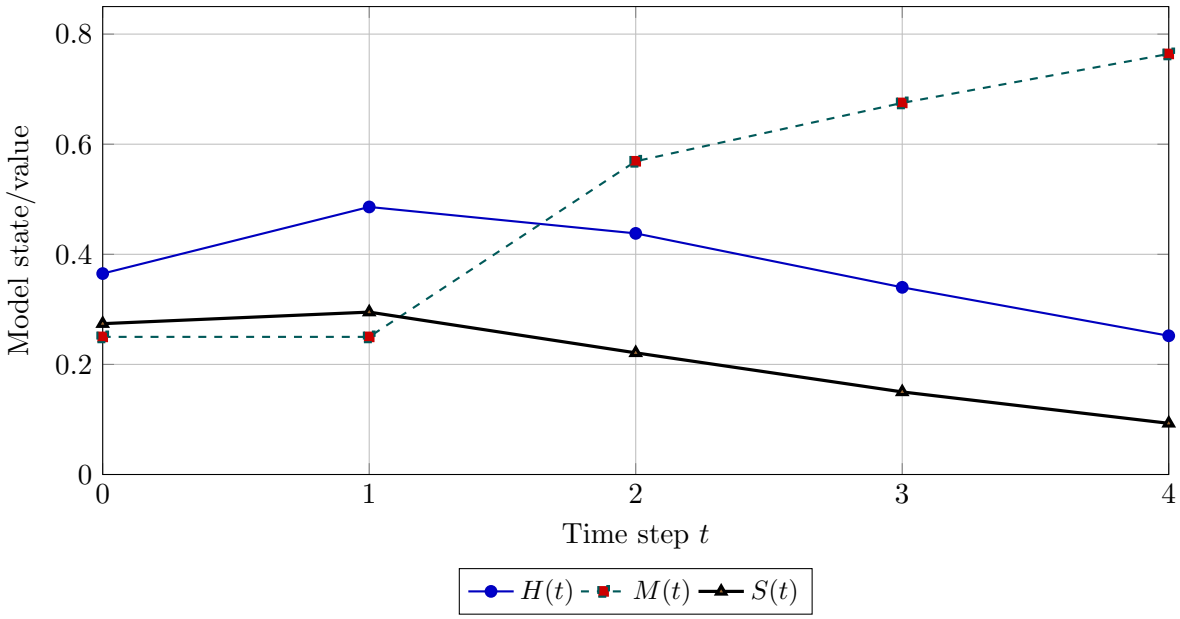


Figure 3: Distinct mechanisms in the worked example: hazard pressure H, effective mitigation M, and persistent unresolved exposure S. The increase in mitigation at t_2 accompanies the decline in subsequent unresolved exposure.

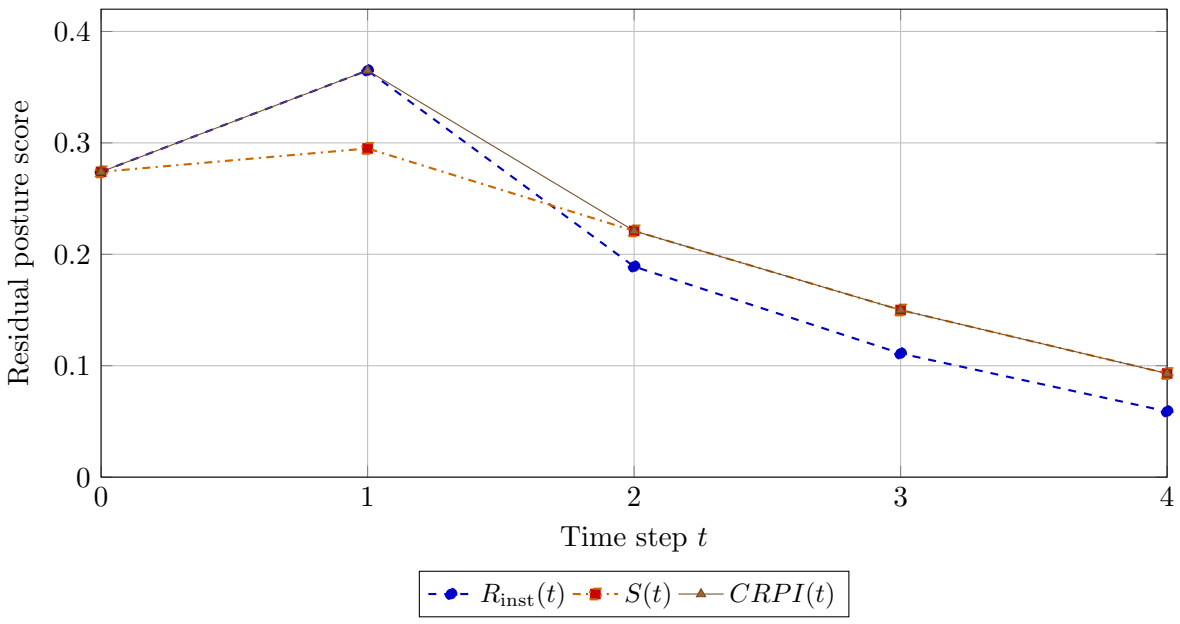


Figure 4: The conservative CRPI envelope (black solid) follows instantaneous residual posture at t_1 and persistent unresolved exposure from t_2 onward. Overlap is intentional and identifies the controlling branch of the maximum.

11 Verification, Validation, and Research Discipline

CRPI is a formally specified operational posture model for which selected mathematical invariants and directional properties are analytically established under stated assumptions. It is not yet empirically validated. The present paper records those boundedness, directional, and persistence properties in Appendix B and exercises them with synthetic checks. Empirical validation requires historical or prospective organizational data and comparison against observed outcomes and simpler baselines.

11.1 Mathematical Verification Objectives

Before empirical deployment, an implementation should reproduce the following properties:

- **Boundedness**: all Tier 2 constructs, D_{eff}, H, M, R_{inst}, S, and $CRPI$ remain in $[0, 1]$ under valid inputs.
- **Directional correctness**: higher TE or consequence increases posture holding other terms constant; stronger effective mitigation lowers R_{inst}; resilience lowers only consequence classes to which ρ_j assigns mitigability.
- **Zero-safe aggregation**: one zero-valued component with positive weight must not force the entire construct to zero; all-zero inputs must return zero and all-one inputs must return one. Components with zero weight have no directional influence by construction.
- **Temporal coherence**: persistent stock accumulates under unresolved residual exposure and dissipates under effective mitigation without leaving $[0, 1]$.
- **Cadence consistency**: changing the reporting interval while preserving the same physical rate parameters and elapsed time should not materially change the modeled persistence trajectory.
- **Decomposability**: every change in $CRPI(t)$ must be traceable through R_{inst}, S, H, M, Tier 2 constructs, and underlying evidence.

11.2 Simulation-Based Stress Testing

Synthetic simulations should include at least:

- **Threat surge**: rapid increase in TE with stable mitigation;
- **Operational improvement**: increase in CM and/or SOE with explicit separation of observability and execution effects;
- **Consequence-specific resilience**: increase domain-applicable projections R_j while varying ρ_j to verify that confidentiality-like irreversible consequence is not incorrectly erased;
- **Delayed remediation**: sustained $R_{\text{inst}} > 0$ with weak M to verify accumulation of S;
- **Recovery**: $R_{\text{inst}} \to 0$ with strong M to verify controlled dissipation of S; and
- **Missing evidence**: remove one entire Tier 2 or governing-path evidence family to verify that coverage gates prevent a high-coverage label.

Parameter sweeps should cover ε_X, m_0, ρ_j, β, κ, α, composition weights, normalization bounds, and reporting cadence. Monte Carlo methods may propagate input uncertainty where defensible input distributions or ranges exist.

The following tests probe sensitivity to input loss, calibration choices, and time aggregation. Their purpose is to reveal model behavior and failure conditions under specified synthetic data. A successful test means the output follows the implemented equation under that constructed condition. It does not mean an organization would experience that outcome or that a particular score is predictive of adverse events.

11.3 Executed Synthetic Stress Experiments

A deterministic study evaluated the existing CRPI equations under the illustrative version 0.1 profile, using seed 20261023. 12,000 complete 15-primitive configurations were independently sampled from Beta$(2, 2)$ on $[0, 1]$. This is an analyst-selected broad input distribution, not a sample of organizations or security incidents. All sampled outputs remained bounded. The median instantaneous residual was 0.1198 and its 90th percentile 0.2154; these synthetic values are not calibrated risk-action thresholds.

Partial observability. The experiments removed observed primitive scores under 10% and 20% missing-completely-at-random mechanisms and a separate risk-correlated omission mechanism. Counterfactual neutral estimates replaced unknown values with 0.5. Independently, the model's monotonic directions supplied mathematically feasible lower and upper residual bounds when unknown input scores were allowed anywhere in $[0, 1]$. The original complete-input result fell within the corresponding bound in every case. These are possible-value ranges, *not* confidence or probability intervals. Table 15 summarizes the experiments.

Table 15: Synthetic residual stress test with 12,000 configurations per mechanism

Missingness	Observed inputs	Median bound width	Bound crosses 0.15	Missed positive cases
10% random	90.1%	0.0487	37.0%	10.8%
20% random	80.0%	0.1387	64.0%	18.5%
Risk-correlated	93.1%	0.0289	29.7%	21.0%

The $R_{\text{inst}} = 0.15$ cutoff is an *illustrative classification boundary*, not a validated control/action threshold. "Missed positives" refers to complete synthetic rows above this boundary for which a neutral fill yielded a value below it. Risk-correlated omissions deliberately favored high external threat, high availability consequence, and weak detection capability. They produced a signed neutral-imputation bias of -0.0090, showing that relatively high input availability need not prevent systematic understatement when the missingness mechanism is informative. At 20% per-field random omission, only 3.18% of rows retained all 15 fields. An incomplete operational assessment must therefore remain explicitly qualified rather than become a falsely precise fully observed score.

Profile and temporal stress. Changing consequence weights reversed 10.5% of 6,000 synthetic pairwise residual rankings. The reference weights are equal thirds; the alternative weights are $(0.6, 0.2, 0.2)$. For a weak-link geometric example $(x_1, x_2) = (0, 0.8)$ with equal weights, the aggregate changed from 0.0273 with $\varepsilon = 0.001$ to 0.1592 with $\varepsilon = 0.05$, and to 0.3915 with $\varepsilon = 0.9$. Profile version and construct calibration materially affect comparability.

For unchanged residual and mitigation, seven one-day exact stock updates agreed with one seven-day update to machine precision. The same statement does *not* justify replacing variable daily inputs with weekly means. In a deliberately extreme sequence containing two days of $(R_{\text{inst}}, M) = (0.8, 0.05)$ and five days of $(0.02, 0.8)$, final stock was 0.0302 with the high-exposure days first and 0.3794 when those days occurred last. Updating once from the common weekly averages yielded 0.1383 in both cases. The difference reflects lost event order, not a failure of the fixed-input cadence theorem. With observed residual 0.05 and mitigation 0.20, a neutral cold start $S_0 = 0.05$ generated a day-one envelope of 0.0557, while unobserved inherited stock $S_0 = 0.85$ yielded 0.7273. Cold-start status must accompany temporal reporting.

Finally, $(H, M) = (0.4, 0.5)$ and $(0.8, 0.75)$ both produce $R_{\text{inst}} = 0.2$ but different subsequent stocks for equal prior states. A lone CRPI scalar cannot explain this distinction. Report H, M, R_{inst}, S, their domain-level inputs, and the evidence qualification separately.

The executable methodology and deterministic results are archived in `scripts/run_stress_tests.py`, `verification/stress_test_results.json`, and `verification/STRESS_TEST_REPORT.md`. These experiments describe formula sensitivity under invented measurements. They do not establish parameter validity, operational prediction, or empirical superiority.

11.4 Illustrative One-Step Sensitivity Check

Using the t_0 worked-example profile with $\bar{\rho} = 0.60$ and $m_0 = 0.25$, the baseline is $D_{\text{eff}} = 0.608$, $H = 0.365$, $M = 0.250$, and $R_{\text{inst}} = 0.274$. Table 16 perturbs selected inputs. This verifies direction only; it is not evidence that the parameter magnitudes are empirically calibrated.

Table 16: One-step sensitivity check for directional correctness

Scenario	D_{eff}	H	M	R_{inst}
Baseline	0.608	0.365	0.250	0.274
Threat surge: $TE = 0.80$	0.608	0.486	0.250	0.365
Consequence rise: $CS = 0.95$	0.722	0.433	0.250	0.325
Domain resilience degradation: $R_j = 0.20$ (special-case $RRAS = 0.20$)	0.704	0.422	0.250	0.317
Monitoring improvement: $CM = 0.75$	0.608	0.365	0.325	0.246
Operations improvement: $SOE = 0.70$	0.608	0.365	0.438	0.205
Balanced mitigation: $CM = 0.75, SOE = 0.70$	0.608	0.365	0.569	0.157
Domain resilience improvement: $R_j = 0.65$ (special-case $RRAS = 0.65$)	0.488	0.293	0.250	0.220

11.5 Empirical Evaluation Boundary

The analytical propositions and synthetic stress tests establish behavior of the specified mathematical model under stated inputs. They do not establish that source observations, normalizers, or calibration parameters measure real organizational cyber-risk posture. The separate operational implementation protocol in Section 12 describes how an organization can instantiate the primitives, conduct independent backtesting, compare alternatives, and assess decision utility.

11.6 Implementation via Statistical Tooling

Statistical environments such as `R` or Python may be used to simulate time series, compute the governing model, perform parameter sweeps, propagate uncertainty, backtest calibration profiles, and reproduce all tables and figures. A publication artifact should include machine-readable metric definitions and executable reference calculations.

11.7 Evaluation Outputs

Validation should produce time-series plots, boundary-condition tests, parameter sensitivity analyses, uncertainty summaries, scenario comparisons, empirical backtest results where data is available, baseline comparisons, and versioned calibration profiles. Until those empirical studies are completed, CRPI should be described as a formally specified operational posture model for which selected mathematical invariants and directional properties are analytically established under stated assumptions. It is not yet empirically validated and should not be described as a validated predictive cyber-risk metric.

12 Organizational Implementation and Empirical Backtesting

The CRPI is an operational measurement design, not an immediately calibrated score that can be copied into every environment. Its mathematical form can be implemented from the preceding specification, but a defensible organizational assessment requires independent decisions about scope, evidence, units, consequence interpretation, parameter identifiability, and available outcomes. The steps below describe a prospective methodology, not a completed organizational deployment or evidence of predictive performance.

12.1 Scope, Evidence and Measurement Design

Begin with a defined unit of analysis, such as a business service within a declared observation period. Record the owner, included applications, asset and identity populations, critical dependencies, data and mission consequences, and the intended decision that a posture assessment could support. Parent-level enterprise scope requires a separately justified evidence boundary; it must not be inferred by averaging child scores.

The next decision is whether the input populations are credible. Reconcile asset/control inventories with independent discovery and identity, cloud, network, endpoint, and dependency sources; deduplicate records, record observation dates, and identify unmatched or disputed entities. Declare both coverage among known sources and unresolved scope completeness. A high declared coverage percentage is not proof that no unobserved population exists. Unknown scope can distort the input values themselves as well as the evidence qualification.

For each of the 15 starter primitives in Table 8, assign an evidence owner, a source-of-record, units, numerator and denominator or an ordinal anchor, event clock, cadence, freshness rule, applicable consequence domain, and missingness behavior. Distinguish not observed from observed zero, no eligible events, expired evidence, withheld evidence, and unverified sources. Have an independent reviewer test both the numerator and denominator of fraction measures. Never use an assertion of source verification as proof of independent authenticity.

12.2 Normalization and Composition Configuration

Each measure must be transformed to $[0, 1]$ using a predeclared mapping with the right directionality. For simple bounded fractions, confirm what population is in the denominator. For event-time measures, declare start and stop events, censored observations, eligible sample sizes, and lower and upper normalization anchors. For consequence severity, document what each impact class actually means and whether numerical spacing between ordinal levels is defensible.

Configure the five Tier 2 constructs using weight vectors that preserve their distinct meanings. Where arithmetic aggregation is chosen, compensating effects between components are allowed; where the regularized geometric operator is used, weak components receive multiplicative penalties without an absorbing zero. These are construct choices that should be reviewed against alternatives and recorded in a named profile. Do not fit a composition weight merely to produce a preferred headline CRPI result.

12.3 Governing Parameter Identification

The model separates composition choices from parameters that assert how capabilities interact. Consequence weights v_j describe the relative role of impact domains. Projection weights $\mathbf{a}_j$ map observed resilience capabilities to those domains, while ρ_j limits the consequences that resilience can change. The independent monitoring baseline m_0 describes response pathways available when continuous monitoring is absent. Regularization ε_X controls zero-safe aggregation and the smoothing coefficient α controls responsiveness to new evidence.

The accumulation and retirement parameters β and κ in the stock equation are especially sensitive. Both have inverse-time units, and the state S is latent: CRPI values computed with assumed rates cannot themselves identify those same rates. Estimation requires separately justified longitudinal evidence of unresolved work, exposure retirement, or another validated state proxy. If the available data cannot identify the two rates, retain them as explicit scenario assumptions and report their sensitivity instead of presenting a fitted value as observed fact.

Table 17: Recommended calibration controls and falsification checks

Family	Evidence or decision basis	Main challenge to test
L, U, ordinal anchors	Independent population and consequence definitions	Unstable denominators, clipping, unequal ordinal intervals
Primitive/construct weights	Defined construct boundaries, elicitation and sensitivity	Double counting, ranking reversal, construct overlap
$\rho_j, \mathbf{a}_j$	Consequence-specific recovery and assurance observations	Credit applied to irreversible or unrelated harms
m_0	Demonstrated manual and noncontinuous response pathways	Overestimated mitigation under weak monitoring
ε_X, α	Weak-component and sampling-cadence sensitivity	Absorbing-zero errors, over-smoothing and delayed reaction
β, κ, S_0	Independent historical state/proxy, time-aligned events	Non-identifiability, missing history, time-aggregation bias
Coverage qualification	Reconciled inventories and independent source trust	False 100% from incomplete known-asset universe

12.4 Historical Replay, Baselines and Holdouts

Evaluate initial configurations against time-ordered observations without allowing future outcomes or later inventory corrections to leak into earlier scores. Preserve input/event ordering within the stock-flow transition; do not substitute a weekly mean for actual daily variation and call the trajectory equivalent. Record cold starts and uncertain history explicitly. When denominator definitions or profiles change, either retain the old series or rerun a labeled historical comparison under the new profile, keeping both source and version lineage.

A credible comparison must include the instantaneous residual without persistence, a simple additive operational index, an aggregation ablation, and at least one independently defined evidence-only baseline. All models must be evaluated on the same eligible service/time rows with matching outcome windows. Comparing scalar magnitudes across indices with different semantics does not measure superiority. The primary outcome must be independently adjudicated, not copied from the controls, exercise results or tickets that generated CRPI primitives.

Keep training and calibration intervals separate from the prospective holdout. Split services by dependency group, and allow sufficient time between partitions to avoid overlapping outcome windows. Fit normalization bounds or any optional probability mapping using training observations alone. Freeze those choices before evaluating withheld outcomes. Report uncertainty that accounts for clustering, missing-data causes, drift, and paired comparisons with simpler baselines. Retain null and negative findings.

To estimate binary event probabilities, a separate fitted and calibrated probability model would be required. The CRPI index does not itself estimate event probability.

12.5 Decision Utility and Operational Governance

The initial deployment should operate in a shadow or research-only mode. Each assessment should expose the five construct scores, consequence domains, resilience projections, hazard, effective mitigation, instantaneous residual, prior and new persistent state, dominant envelope branch, profile/version identifier, observation period, evidence lineage, and coverage qualification. Do not automate consequential decisions from an unqualified scalar or a threshold copied from the synthetic examples.

If a decision use case can be defined, predeclare the actual intervention being supported, such as allocating a fixed number of independent security reviews, and compare the outcome and cost of that ranking decision with simpler alternatives. Recalibration requires a documented trigger, sensitivity assessment, independent review, version freeze, rollback path, and clear separation of the old and new series. If independent evidence does not demonstrate added measurement reliability or practical decision value, the appropriate result is to limit or replace the provisional proxy or model claim.

This process is a study protocol rather than an authorization to collect organizational records. Appropriate organizational consent, legal/data-use permissions, privacy handling, and independent ethics determination where applicable must precede non-synthetic research data collection.

13 Related Work and Positioning

Prior cyber-risk scholarship spans governance, financial quantification, vulnerability prioritization, probabilistic dynamic risk assessment, architecture-aware analysis, resilience, and security measurement.

Governance frameworks including NIST CSF, NIST SP 800-30/39, NIST IR 8286, ISO/IEC 27001, ISO/IEC 27005, ISO 31000, and CIS Controls are strong at defining risk processes, accountability, control expectations, and assurance lifecycles [1, 2, 3, 5, 6, 7, 8, 9]. They generally do not prescribe one continuously updated, bounded organizational posture equation. CRPI is positioned as a framework-compatible mathematical layer above such governance systems, not a replacement for them.

Quantitative cyber-risk approaches including FAIR and economics-oriented literature provide rigorous reasoning about loss-event frequency, magnitude, uncertainty, and investment trade-offs [10, 19, 17, 18, 20]. CRPI addresses a different question. It does not estimate financial loss or breach probability; it targets a continuously updateable operational posture signal that may serve as evidence for downstream scenario or financial analysis.

Vulnerability and exploitability approaches including CVSS, EPSS, CISA KEV, and SSVC provide actionable signals at vulnerability granularity [11, 12, 14, 15, 16]. CRPI treats such values as possible TE evidence rather than complete organization-level posture measures.

Dynamic cyber-risk research already includes Bayesian networks, Markov models, attack graphs, simulation, and intelligence-driven approaches. Cheimonidis and Rantos propose a proactive dynamic methodology using open-source evidence, EPSS, and a Bayesian network to estimate attack success probability in a SCADA use case [29]. More recent DRIMAX work combines intelligence, context-aware scoring, attack paths, and dynamic assessment in critical environments [30]. These approaches reinforce the importance of time-varying evidence and contextual dependencies while clarifying CRPI's narrower contribution: CRPI is not a competing attack-probability engine. It is a telemetry-compatible, framework-agnostic, decomposable operational posture index that separates instantaneous residual posture from persistent unresolved exposure and supplies a governed measurement interface.

Architecture and critical-infrastructure research further emphasizes dependency structure, cascading impact, and rigorous validation [21, 22]. Resilience and survivability literature emphasizes continuity, recoverability, dependency assurance, and security categorization [4, 3, 6, 22]. CRPI incorporates resilience explicitly but constrains its effect through consequence-specific mitigability factors rather than assuming that recoverability can erase all harm.

Security measurement research, including NIST SP 800-55, contributes disciplined methods for measure definition, normalization governance, data quality, and repeatable measurement operations [23, 24]. CRPI adopts this measurement orientation and adds a stable state model, explicit zero-safe aggregation, evidence coverage gates, and temporal persistence. The intended contribution is therefore the combination of operational telemetry compatibility, mathematical decomposability, governed adaptation, and stateful posture representation in a single bounded index.

14 Assumptions and Limitations

The CRPI is mathematically specified but not assumption-free.

First, normalization quality depends on calibration quality. Poorly chosen bounds can degrade interpretability even if equation implementation is correct.

Second, several inputs are proxies rather than direct observables. Exploitability, derived sensitivity, threat relevance, dependency trustworthiness, and resilience applicability may rely on modeled or aggregated evidence rather than ground truth.

Third, composition and calibration parameters remain judgment-bearing quantities. Regularization ε_X, consequence weights v_j, resilience mitigability ρ_j, the observability-independent baseline m_0, smoothing, and persistence rates must be empirically reviewed. CRPI reduces subjectivity by

localizing and governing these choices; it does not eliminate subjectivity.

Fourth, the consequence-domain model assumes that a finite governed set $\mathcal{D}$ can adequately represent the material forms of harm in scope. Poor domain definition or double-counting across domains can invalidate D_{eff}.

Fifth, evidence completeness materially affects interpretation. Aggregate coverage alone is insufficient, which is why the model reports $C_{\min}$ as a hard diagnostic. Even full coverage does not imply low uncertainty if the observed measures themselves are noisy or weak proxies.

Sixth, the stock-flow model is an abstraction. The functional form in Equation 28 provides desirable bounded behavior, but β and κ require empirical calibration. Different organizations may exhibit different accumulation and remediation dynamics.

Seventh, the CRPI measures operational posture rather than direct financial loss or breach probability. It can inform enterprise risk management but does not replace scenario-based loss quantification [17, 18].

Eighth, AI-era public data cannot isolate an AI treatment effect. CVE publication volume is influenced by reporting-system growth, including expansion of the CNA ecosystem, and KEV is a curated catalog with its own inclusion and collection processes. The public illustration is therefore descriptive and mechanistic: it demonstrates how changing signals propagate through CRPI, not why those signals changed.

14.1 Threats to Validity

The primary threats to validity are measurement validity, construct validity, parameter-identification validity, external validity, and incentive distortion. Measurement validity is threatened by incomplete, stale, or biased telemetry. Construct validity is threatened when evidence is assigned to the wrong Tier 2 or consequence domain, when a metric is double-counted, or when an aggregation operator does not match construct semantics. Parameter-identification validity is threatened when values are chosen from convenience rather than data or defensible expert elicitation. External validity is threatened when calibration profiles are transferred between organizations or environments without recalibration. Incentive distortion is threatened when teams optimize visible sub-metrics instead of reducing actual exposure.

The CRPI mitigates these risks through explicit evidence ownership, coverage gates, versioned definitions and parameters, uncertainty reporting, outcome-informed recalibration, and end-to-end decomposition. These mechanisms make model risk observable and governable; they do not eliminate it.

15 Governance, Adaptation, and Longevity

For a posture model to remain useful over long time horizons, adaptation must be explicit, controlled, and auditable. CRPI therefore separates stable equation form from versioned measurement and calibration profiles.

At a minimum, implementations should enforce:

- **Explicit Evidence Ownership**: each primitive measurement is assigned to one primary construct/domain unless a documented transformation prevents double-counting;
- **Versioned Metric Definitions**: sub-metric definitions, evidence sources, normalization functions, and consequence-domain mappings are versioned;

- **Versioned Parameters**: α, ε_X, v_j, ρ_j, m_0, β, κ, coverage gates, and time units are managed as named calibration profiles;
- **Outcome-Informed Recalibration**: incidents, near misses, recovery performance, validation failures, and distribution shifts inform recalibration;
- **Sensitivity Review**: proposed parameter changes are evaluated against representative scenarios and historical data before adoption;
- **End-to-End Traceability**: every CRPI value can be decomposed into R_{inst}, S, H, M, D_{eff}, Tier 2 constructs, domain components, and underlying measurements; and
- **Comparability Controls**: historical outputs remain tied to their calibration profile, and recalculated historical series are explicitly labeled.

These controls allow telemetry, organizational context, technologies, and threat environments to evolve without silently changing what the governing model means. The goal is controlled adaptation rather than a claim of universal coefficients [3, 5, 8]. This is particularly important when AI-related or other rapid environmental changes shorten the useful life of historical calibration assumptions [25, 27].

16 Reproducibility and Artifact Availability

The accompanying public artifact repository supplies a provisional 15-input operational measurement specification and reproducible synthetic missingness, calibration-profile, and temporal-order experiments. The accompanying repository contains the manuscript source, an executable dependency-light reference implementation of the governing equations, a versioned illustrative calibration profile, synthetic non-sensitive scenario data, unit and invariant tests, reference calculations, and scripts that reproduce the scenario tables and decomposed outputs. The artifact is intended to make equation interpretation, parameterization, coverage qualification, and use-case behavior inspectable. It does not contain empirical organizational data and does not claim empirical validation.

17 Conclusion

This work presents the Cyber Risk Posture Index as a telemetry-compatible, framework-agnostic, decomposable operational posture index for representing cyber risk over time. Its contribution is not a universal law of cyber risk or a replacement for probability- or loss-based quantification. It is a stable mathematical interface that separates current hazard, effective mitigation, instantaneous residual posture, and persistent unresolved exposure while preserving traceability to the evidence that produced those values.

The revised governing model addresses several important failure modes of simpler scoring systems. Regularized geometric aggregation prevents accidental absorbing zeros; consequence-specific resilience avoids treating recoverability as universally capable of erasing impact; observability modulates operational execution rather than acting as remediation by itself; and the bounded stock-flow equation gives persistent exposure an explicit accumulation and dissipation mechanism. The top-level CRPI value remains a conservative output envelope rather than a breach probability.

The model also makes its remaining judgment explicit. Composition weights, regularization, resilience mitigability, observability baselines, normalization, and temporal rates are governed calibration parameters, not hidden preferences. The research claim is therefore narrower but more defensible: CRPI is a formally specified operational posture model for which selected mathematical invariants and directional properties are analytically established under stated assumptions. It is not yet empirically validated. CRPI constrains, localizes, and exposes subjectivity through mathematically specified and testable relationships. Empirical backtesting, baseline comparison, and parameter identification remain required before any stronger empirical claim is warranted.

A Notation and Definitions

Table 18 summarizes the principal variables used throughout the paper.

Table 18: Core notation and variable definitions

Symbol	Meaning
$\mathbf{a}_j$	Non-negative weight vector that projects shared resilience primitives onto consequence domain j, with weights summing to one.
α	EWMA smoothing parameter that controls the weight assigned to the current observation.
β	Accumulation-rate parameter for unresolved exposure, measured in inverse time.
$\bar{C}_{\mathrm{CRPI}}(t)$	Arithmetic mean of the Tier 2 evidence-coverage scores at time t.
$C_{D_j}(t)$	Evidence-coverage score supporting the consequence-domain score $D_j(t)$.
$C_{\mathrm{gov}}(t)$	Minimum evidence-coverage gate over the constructs and consequence-specific paths used by CRPI.
$C_{\min}(t)$	Minimum evidence-coverage score across the Tier 2 constructs at time t.
$C_{R_j}(t)$	Evidence-coverage score supporting the resilience projection $R_j(t)$.
$C_X(t)$	Evidence-coverage score for construct X at time t.
$CM(t)$	Continuous Monitoring score measuring how completely and reliably security-relevant state is observed and verified, bounded in $[0, 1]$.
$CRPI(t)$	Conservative operational cyber-risk output envelope formed from instantaneous residual posture and persistent unresolved exposure.
$CS(t)$	Consequence Severity aggregate representing the weighted severity of governed consequence domains, bounded in $[0, 1]$.
$D_{\mathrm{eff}}(t)$	Consequence remaining after domain-specific resilience adjustment and used in the hazard calculation.
$D_j(t)$	Consequence score for governed consequence domain j at time t, bounded in $[0, 1]$.
Δt	Elapsed time between updates to the persistent-exposure state.
ε_X	Positive regularization parameter that makes geometric aggregation zero-safe for construct X.
$H(t)$	Hazard pressure after threat exposure is combined with resilience-adjusted consequence.

Symbol	Meaning
κ	Dissipation-rate parameter governing how quickly persistent exposure decays, measured in inverse time.
L, U	Lower and upper raw-value bounds used to normalize a sub-metric.
$M(t)$	Effective mitigation capacity after continuous monitoring modulates operational execution.
m_0	Baseline fraction of operational effectiveness that does not depend on continuous monitoring.
$R_{\text{inst}}(t)$	Instantaneous residual posture remaining after current hazard pressure is reduced by effective mitigation.
$R_j(t)$	Resilience capability projected onto consequence domain j, bounded in $[0, 1]$.
$RRAS(t)$	Resilience and Risk Assurance Score summarizing structural resilience and dependency assurance, bounded in $[0, 1]$.
ρ_j	Maximum fraction of consequence domain j that its resilience projection can mitigate, bounded in $[0, 1]$.
$S(t)$	Bounded stock of persistent unresolved residual exposure at time t.
$TE(t)$	Threat Exposure score representing adversary pressure and exposure conditions, bounded in $[0, 1]$.
$SOE(t)$	Security Operations Effectiveness representing operational execution and mitigation capability, bounded in $[0, 1]$.
v_j	Consequence-domain composition weight for domain j, with $v_j \geq 0$ and $\sum_j v_j = 1$.
$\widetilde{x}$	Normalized raw or derived sub-metric bounded in $[0, 1]$.

B Appendix: Analytical Properties

This appendix records the principal mathematical properties of the governing model under its stated assumptions. Let all normalized inputs lie in $[0, 1]$, let all composition weights be non-negative and sum to one, let $\varepsilon \in (0, 1)$, and let $\beta, \kappa \geq 0$. The results concern boundedness, direction, and temporal behavior; they do not establish empirical validity.

Proposition 1 (Regularized aggregation). The operator $\mathcal{G}_\varepsilon$ maps $[0, 1]^n$ to $[0, 1]$, preserves the all-zero and all-one endpoints, and is non-decreasing in each input with positive weight.

Proof. Write $y_i = \varepsilon + (1 - \varepsilon)x_i$. Then $y_i \in [\varepsilon, 1]$ and the weighted geometric product $P = \prod_i y_i^{w_i}$ lies in $[\varepsilon, 1]$ because $\sum_i w_i = 1$. Therefore $(P - \varepsilon)/(1 - \varepsilon) \in [0, 1]$. If all $x_i = 0$, then $P = \varepsilon$ and the result is zero; if all $x_i = 1$, then $P = 1$ and the result is one. For $w_k > 0$,

$$\frac{\partial \mathcal{G}_\varepsilon}{\partial x_k} = \frac{w_k P}{\varepsilon + (1 - \varepsilon)x_k} \geq 0, \tag{44}$$

which proves monotonicity. □

Proposition 2 (Bounded consequence, hazard, and mitigation). Under valid inputs, D_{eff}, H, M, and R_{inst} all lie in $[0, 1]$.

Proof. Each factor $1-\rho_j R_j$ lies in $[0,1]$, so each term $v_j D_j(1-\rho_j R_j)$ lies in $[0, v_j]$. Summing gives $D_{\text{eff}} \in [0,1]$. The product $H = TE\, D_{\text{eff}}$ is therefore bounded. The quantity $m_0 + (1-m_0)CM$ is a convex combination of m_0 and one, so $M = SOE[m_0 + (1-m_0)CM] \in [0,1]$. Finally, $R_{\text{inst}} = H(1-M) \in [0,1]$. □

Proposition 3 (Persistence invariance and equilibrium). For $S(t_0) \in [0,1]$, the stock-flow model preserves $S(t) \in [0,1]$. Under constant R_{inst} and M, if $q = \beta R_{\text{inst}} + \kappa M > 0$, the unique equilibrium is $S^* = \beta R_{\text{inst}}/q$ and the exact solution converges to it.

Proof. Let $a = \beta R_{\text{inst}}$ and $b = \kappa M$. At $S = 0$, the vector field is $a \geq 0$; at $S = 1$, it is $-b \leq 0$. Thus the interval boundaries point inward. When $q = a + b > 0$, the ODE is $\dot{S} = a - qS$, whose unique equilibrium is $S^* = a/q$. The interval update is

$$S(t + \Delta t) = S^* + [S(t) - S^*]e^{-q\Delta t}. \tag{45}$$

It is a convex combination of $S(t)$ and S^*, both in $[0,1]$, and the exponential factor tends to zero as time increases. If $q = 0$, then $a = b = 0$ and the stock remains constant. □

Proposition 4 (Cadence consistency). For constant interval inputs, applying the exact update over adjacent intervals is equivalent to applying it once over their combined duration.

Proof. With constant S^* and q, applying durations Δt_1 and Δt_2 gives

$$S^* + \left[S^* + (S - S^*)e^{-q\Delta t_1} - S^*\right] e^{-q\Delta t_2} = S^* + (S - S^*)e^{-q(\Delta t_1 + \Delta t_2)}, \tag{46}$$

which is the single combined-interval update. □

Proposition 5 (Directional sensitivity). Holding other valid terms fixed, R_{inst} is non-decreasing in TE and each D_j, and non-increasing in each domain projection R_j, SOE, and CM.

Proof. The partial derivatives of effective consequence with respect to D_j and R_j are $v_j(1 - \rho_j R_j) \geq 0$ and $-v_j D_j \rho_j \leq 0$, respectively. Since $H = TE\, D_{\text{eff}}$, its direction follows. Also,

$$\frac{\partial M}{\partial SOE} = m_0 + (1-m_0)CM \geq 0, \qquad \frac{\partial M}{\partial CM} = SOE(1-m_0) \geq 0. \tag{47}$$

Because $R_{\text{inst}} = H(1-M)$, increasing either mitigation input cannot increase instantaneous residual posture. Strict inequalities require non-degenerate values for the relevant multiplicative factors. □

Proposition 6 (CRPI boundedness and envelope switching). If $R_{\text{inst}}(t), S(t) \in [0,1]$, then $CRPI(t) = \max(R_{\text{inst}}(t), S(t)) \in [0,1]$. The envelope is non-decreasing in each component and is non-differentiable on the switching boundary $R_{\text{inst}}(t) = S(t)$. The boundary is an output characteristic, not a defect; gradient-based calibration should operate on the underlying differentiable components or use an explicitly governed smooth surrogate rather than silently treating the maximum as differentiable.

Proof. The maximum of two values in $[0,1]$ is in $[0,1]$. If either component increases while the other is held fixed, their maximum cannot decrease. The two affine branches have different gradients at $R_{\text{inst}} = S$, so the maximum is not differentiable there. □

C Appendix: Sub-metric Measurement Detail

The detailed row-level measurement specifications referenced in the Measurement Specification section should be provided as supplementary material or an implementation artifact. Formula expressions should use the common normalization conventions defined there, where $N_+(x)$ denotes higher-is-better normalization and $N_-(x)$ denotes lower-is-better normalization.

C.1 Supplementary Measurement Tables

For publication, the full row-level sub-metric measurement tables should be maintained as supplementary material or an implementation artifact rather than forced into the main manuscript. This avoids reducing readability of the main article while preserving reproducibility for implementers. The supplementary measurement artifact should include, for each baseline sub-metric, the following fields: construct owner, formula or normalization direction, evidence source, expected cadence, missing-data policy, normalization bounds, calibration profile, and boundary notes.

The baseline supplementary table set should contain the following construct and consequence groups:

- **CM**: telemetry coverage, freshness, integrity, audit-log coverage, control-state drift detection, control update cadence, and procedure freshness.
- **CS**: consequence-domain mappings for confidentiality/sensitivity, integrity/mission, and availability/continuity, including direct and derived sensitivity and business criticality.
- **TE**: attack-surface exposure and exploitability signal; privilege/identity exposure; threat-intelligence relevance; and control-bypass potential.
- **SOE**: MTTD, MTTR, MTTP, change failure rate, deployment regularity, preventive quality coverage, and supply-chain execution assurance.
- **RRAS**: recoverability, redundancy/failover, diversity, dependency transparency, dependency trust, and assured change integrity.

A publication artifact should additionally include machine-readable definitions for $N_+(x)$ and $N_-(x)$, the raw-to-normalized bounds (L, U), the governing parameters $(\varepsilon_X, v_j, \rho_j, \mathbf{a}_j, m_0, \beta, \kappa, \alpha)$, the declared time unit, the domain-projection mappings used to construct R_j, and a version identifier for each calibration profile. This preserves the paper's traceability requirements without requiring dense implementation tables to consume main-paper space.

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